

A \$12 million battery. A \$57 million project.

Hyde Park BESS (Project 21334): where the cost of a utility-owned, distribution-connected battery storage station actually sits, and how it compares to industry benchmarks

Cost Estimating Center of Excellence | Estimate basis: ESF Total Rev 3, 24 Sep 2026 | 5 Oct 2026

Internal. Contains vendor-confidential pricing.

THE PROMISE

By the end of this deck you can answer one question in one breath

“Why is a \$12 million battery a \$57 million project?” Because a vendor prices the battery. We build the station, the site, the feeder protection and the siting case, and we carry the risk and the cost of money.

\$12.1M

Vendor fixed-price contract

Nomad cover agreement, 10 Mar 2026.
12 Voyager LFP units, PCS, BMS, fire protection, 10-yr warranty. \$605/kWh, \$1,211/kW.

\$57.2M

Project 21334 approval level

ESF Total Rev 3. Station 360 \$50.0M plus Station 496 protection \$7.3M. \$2,862/kWh, \$5,725/kW. Conceptual, -25%/+50%.

21%

Battery share of the total

Add the D-Line circuit extension (24211, \$4.8M conceptual) and the share drops to 20%. Four dollars of every five are not the battery.

Three sections: where the money sits (thirteen layers, unit costs, benchmarks, six price curves); how the cost changed (2021 to today); can the number be defended (red team, stress tests, what must close).

Against 35 public projects its own size: the battery installed sits at the 57% percentile, the station complete at the 77% (68% among the 25 utility-owned), and the approval level above every member. The gap is the station, the interconnection and the loaders, not the battery.

What a vendor quote contains, and what it leaves for the utility

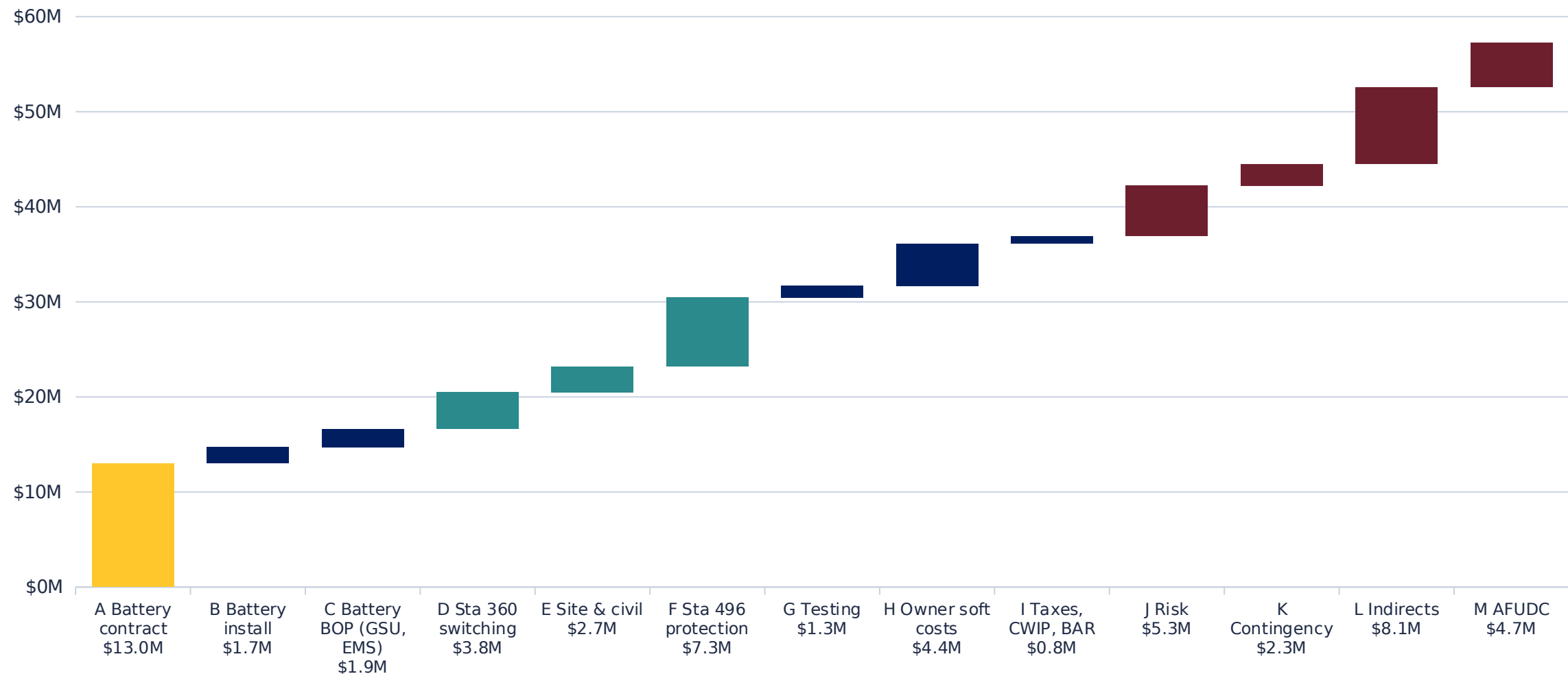
Inside the \$12.1M Nomad price

- 12 containerized LFP units, 10 MW / 20 MWh, 12 MVA
- Inverters (PCS), BMS, HVAC, fire detection and suppression inside the container
- Spares \$125K, shipping \$90K, remote support \$210K
- 10-year warranty; years 3-7 extension \$798K
- Field assembly supervision, commissioning, training: \$246K
- Fixed until Final Acceptance; LDs capped at 10%

Outside it: \$45.1M, 79% of the project

- **Station 360**, \$3.8M direct: four 15 kV breaker skids, GSUs, SSVT, metering, cable, duct bank, ground grid
- **Site**, \$2.7M direct: a 200 ft x 200 ft platform on a closed landfill, access road, retaining wall
- **Station 496**, \$7.3M: 20 relays, transfer trip, RTU work so four feeders can accept 2.5 MW each
- **Owner scope**, \$10.1M: install labor, GSUs, EMS, testing, engineering, Article 80 siting, outreach, PM, taxes
- **Station 360 risk, contingency, indirects, AFUDC**, \$20.3M: 35% of the total (Station 496 carries its own inside its \$7.3M)

Thirteen layers between the battery contract and the approval level



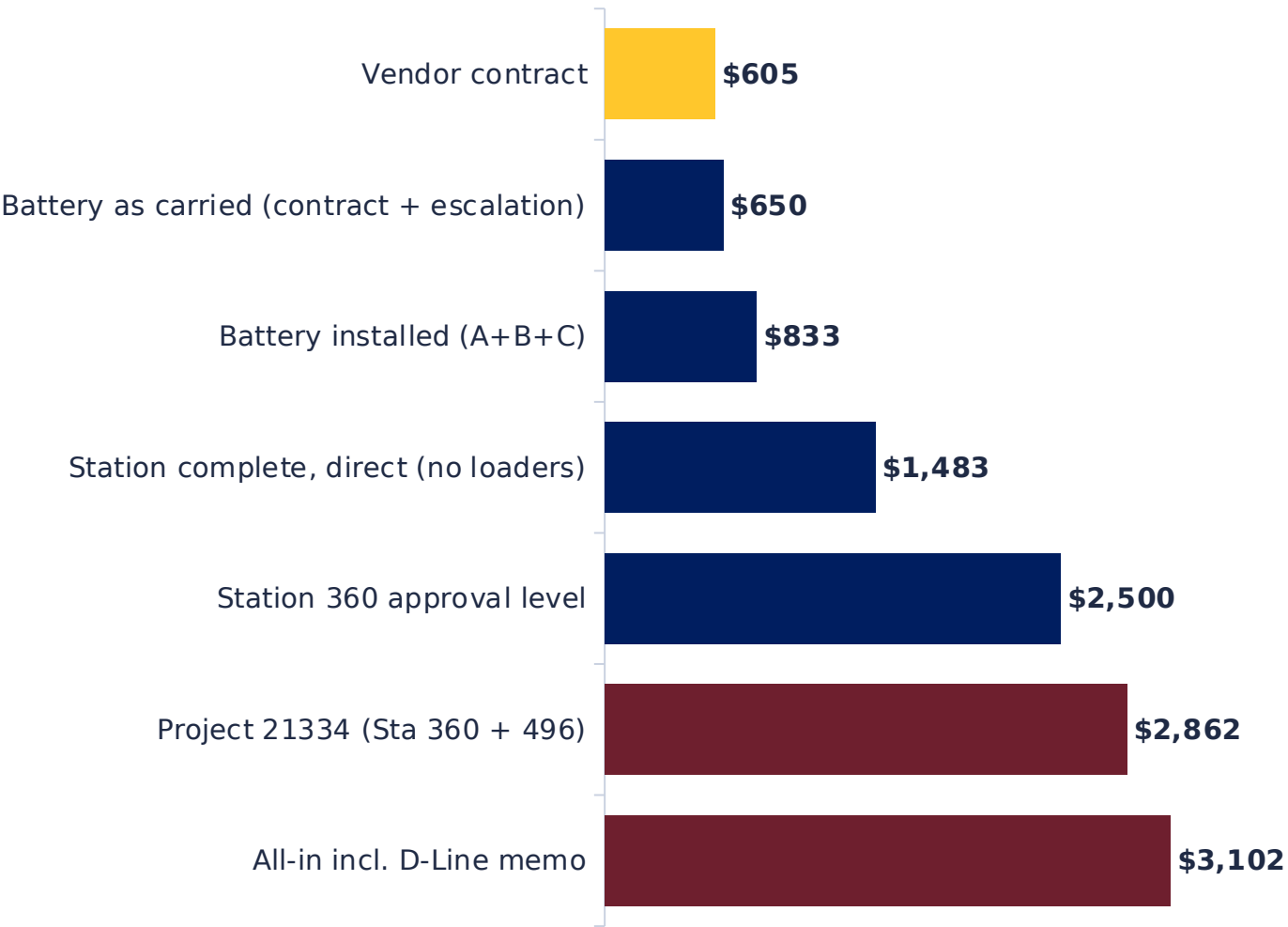
Gold = battery contract as carried in Rev 3 (\$12.1M contract plus \$1.1M escalation on the unpaid balance; see slide 23). Teal = station, site and feeder protection. Burgundy = risk, contingency, indirects, AFUDC. Total \$57.2M.
EstimatingCoE | Eversource Energy | Stewards of Affordability

Where the \$57.2M sits when risk, contingency, indirects and AFUDC are spread

Layer	Direct	Loaded	Share
Battery supply contract (Nomad)	\$13.0M	\$21.9M	38%
Battery installation & integration (owner scope)	\$1.7M	\$2.9M	5%
Battery balance-of-plant (GSUs, EMS/comms, aux power)	\$1.9M	\$3.3M	6%
Station 360 13.8 kV switching & interconnection facilities	\$3.8M	\$6.5M	11%
Site development & civil (former landfill site)	\$2.7M	\$4.5M	8%
Station 496 feeder protection upgrades (interconnection)	\$7.3M	\$7.3M	13%
Testing & commissioning	\$1.3M	\$2.1M	4%
Owner soft costs (engineering, siting, environmental, outreach, PM, land)	\$4.4M	\$7.4M	13%
Taxes, CWIP, builders-risk insurance	\$0.8M	\$1.3M	2%
Total, project 21334 (directs incl. \$20.3M of loaders J-M)	\$57.2M	\$57.2M	100%

Battery-related layers A+B+C loaded: \$28.1M (49%). Station, site and feeder interconnection D+E+F loaded: \$18.3M (32%). Owner soft costs, testing and taxes: \$10.9M.

The same project is \$605/kWh or \$2,862/kWh depending on where you draw the line

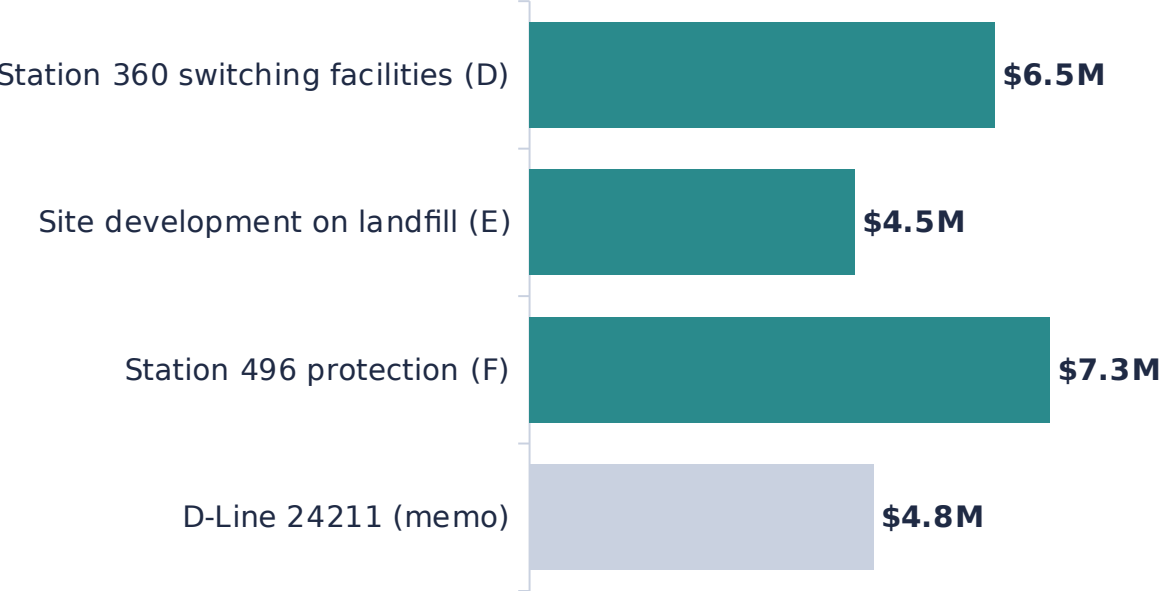


Boundary	\$/kW	\$/kWh	x
Vendor contract	\$1,211	\$605	1.0x
Battery installed	\$1,666	\$833	1.4x
Sta 360 approval	\$4,999	\$2,500	4.1x
Project 21334	\$5,725	\$2,862	4.7x

Incl. D-Line \$6,204 \$3,102 5.1x

A benchmark is only meaningful against the row that matches its definition. Pack and DC-block prices belong beside the top bar; NREL and EIA installed costs beside 'battery installed' or 'station complete'; nothing published includes a utility's own indirects and AFUDC.

Station and interconnection: \$18.3M loaded, none of it in a vendor quote

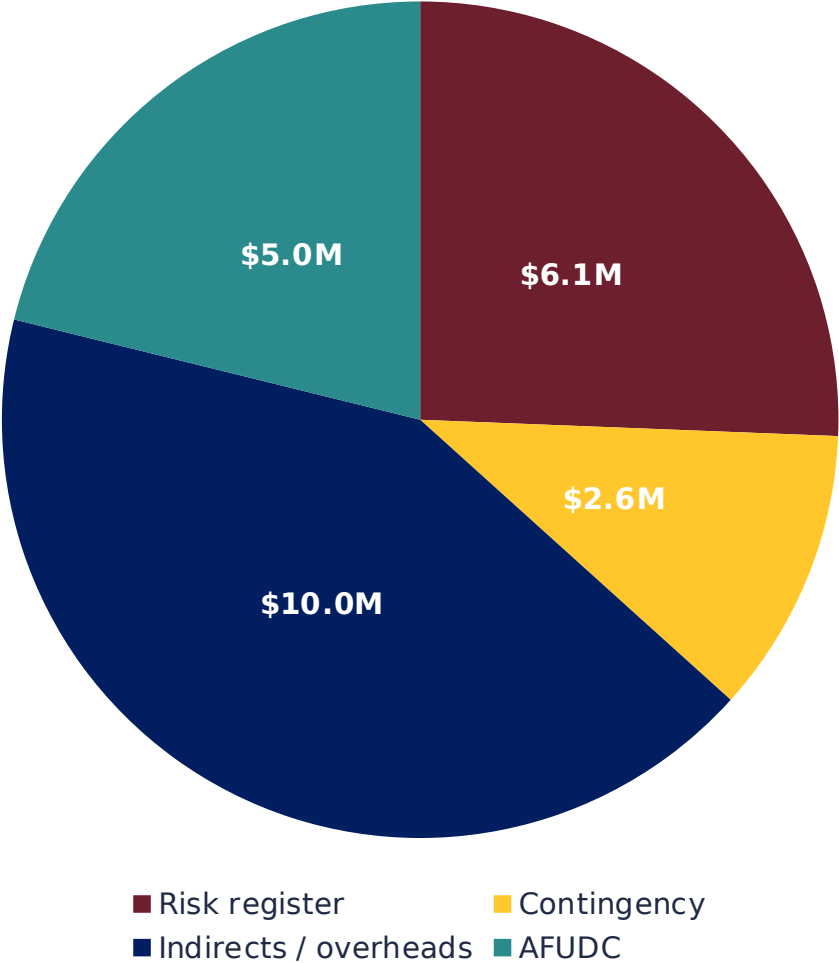


- **Four Siemens SDV7 breaker skids at \$300K each** (relays, PTs, DC battery, cabinet on the skid), disconnects, four metering transclosures, station service, riser poles, 15 kV cable and 1,240 ft of duct bank.
- **Station 496:** 20 SEL relays replacing SEL-551 and REF610 units on four feeders, RTU and RTAC points, transfer trip over dark fiber; \$1.8M of that is testing.
- **Site:** 2.5 acres cleared, 760 ft road, 2,520 CY cut, imported fill, 14 ft retaining wall, ground grid, fence, lighting; Haugland's platform take-off is larger than ours and is unreconciled.
- **D-Line (24211):** 5,465 ft feeder extension, three manholes, 16,395 ft of 3-700 cable; \$4.8M is a 2023 conceptual figure.

\$18.3M
Loaded, in 21334
32% of the approval level

\$23.1M
Incl. D-Line memo
37% of the all-in view

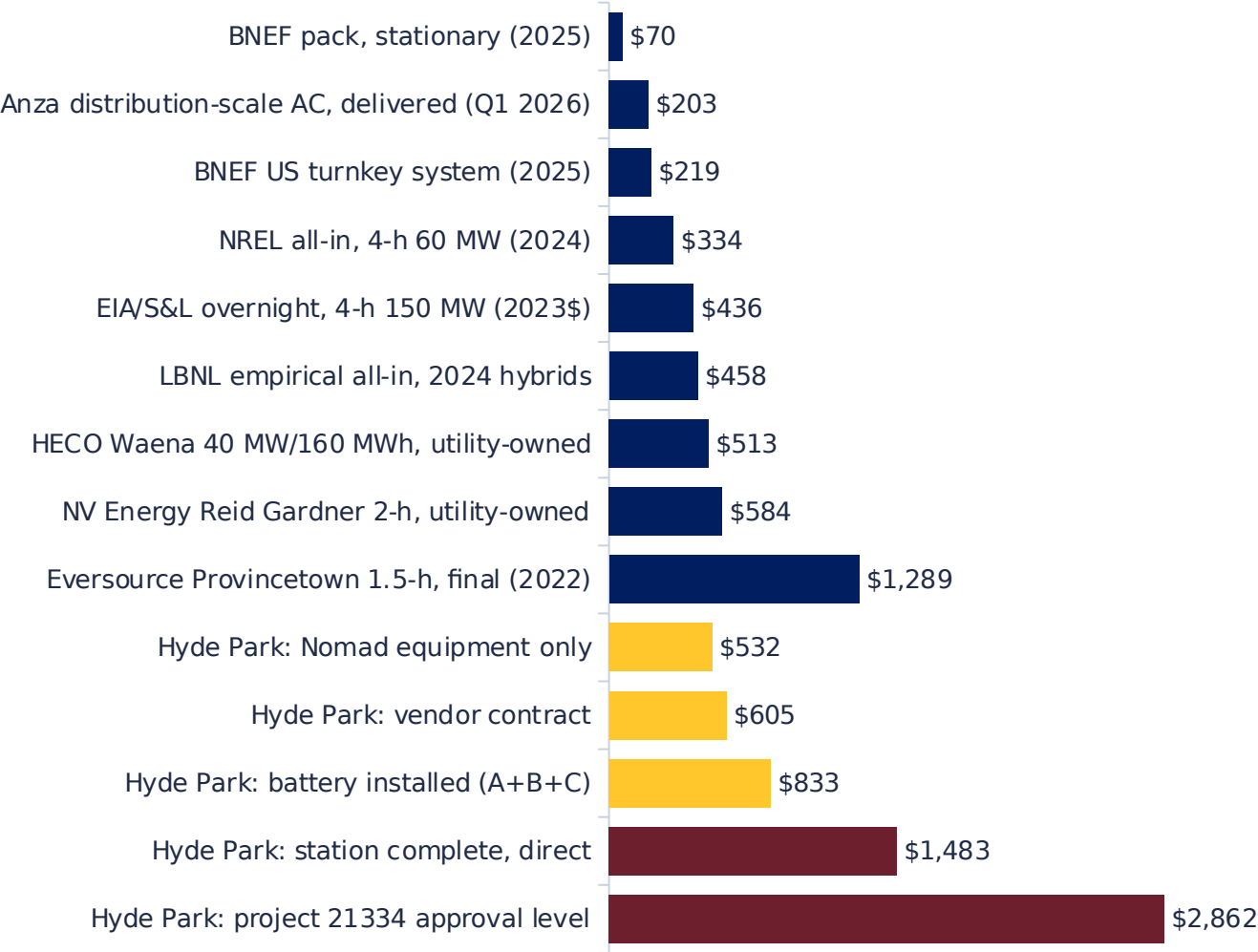
Risk, contingency, indirects and AFUDC are 41% of the approval level



Station 360 risk item	\$K
Material cost, tariffs	1,000
Design / scope change orders	800
Labor cost, shortage, overtime	800
Unforeseen UG obstructions (landfill)	450
Environmental (DEP, landfill closure docs)	300
Soil contamination remediation	300
Station and P&C engineering changes	600
Weather, siting delay, outreach, BAR deductibles	1,050

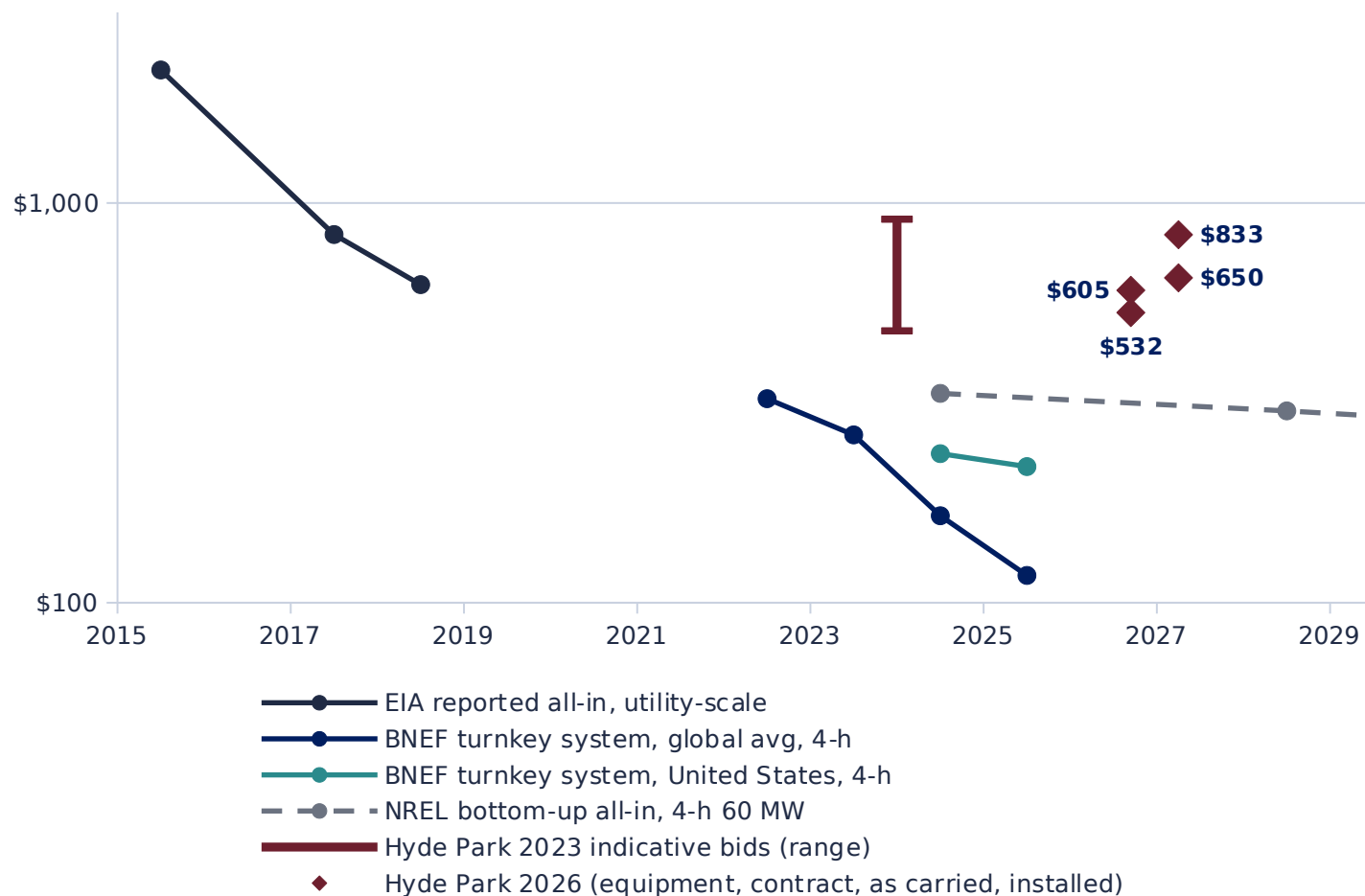
Risk plus contingency is 17.9% of the approval level before loaders, inside the 15-18% lead-review guidance. Indirects at 17.4% and AFUDC at 8.8% follow the EMA station template on a spend curve that runs from July 2021 to December 2028. Haugland's Attachment 4-F carries overlapping exposures (geotech \$0.6M-\$3.5M, fault-study equipment change \$0.25M-\$1.8M) that have not been reconciled with this register.

The battery is at market; the project is above it for reasons benchmarks exclude



- **Battery at market.** Nomad's \$532/kWh equipment price sits between BNEF's US turnkey (\$219, 4-h, 100 MW class) and the disclosed utility-owned projects (\$513-584). For a 12-unit, 2-h, FEOC-era order in New England that is a fair price.
- **Installed battery (A+B+C) at \$833/kWh** is above every national all-in index because those indices are four-hour and 60-150 MW. NREL's own cost structure (pack \$/kWh x hours + BOS \$/kW) puts a 2-h system at roughly 1.3-1.5x the 4-h \$/kWh.
- **The project at \$2,862/kWh** is not a battery benchmark at all. It carries a switching station, feeder protection at Station 496, a landfill platform, Article 80 siting, indirects and AFUDC. The closest published analogue is Provincetown at \$1,289/kWh (2022 dollars, no separate station, existing site).
- **No published index includes a utility's indirects or AFUDC.** Strip J-M and Hyde Park is \$1,483/kWh direct; strip the station and site layers D and E as well and it is \$1,156/kWh.

Battery installed-cost curve: the Nomad price sits on it once duration and size are adjusted



■ **The curve is a 4-h, 60-150 MW, real-dollar curve.** EIA's reported all-in fell from \$2,152 (2015) to \$625 (2018); BNEF turnkey from \$324 (2022) to \$117 global and \$219 US (2025); NREL's 4-h bottom-up base is \$334 (2024), \$247 by 2035.

■ **Hyde Park is 2-h, 10 MW, nominal.** NREL's structure puts a 2-h system at 1.3-1.5x the 4-h \$/kWh; the size elasticity in Appendix B adds 1.5x at 20 MWh. Adjusted US turnkey: \$440-\$505. Nomad equipment: \$532.

■ **Install is the layer the curve never shows.** Owner-scope installation, integration and balance of plant (B+C) add \$183/kWh to the \$650 battery line: \$833 installed. BNEF's install sits inside a developer's EPC; ours is Eversource scope at New England rates.

■ **The 2026 turn.** Lithium carbonate roughly doubled year on year by August 2026 and tariffs lifted delivered US prices. The March 2026 fixed price sits below where a 2027 re-procurement would land.

Plotted as published: EIA-860 nominal; BNEF real survey-year; NREL 2024\$ (2028 interpolated between the 2024 base and the 2035 mid case). 2023 bar = five indicative bids, \$479-\$911/kWh, differing scopes. Retrieved from search excerpts; verify before external use.

Escalation is already in the number: \$2.9M at template rates, \$1.1M of it on a fixed price

Where the escalation sits (Rev 3, included in the directs)

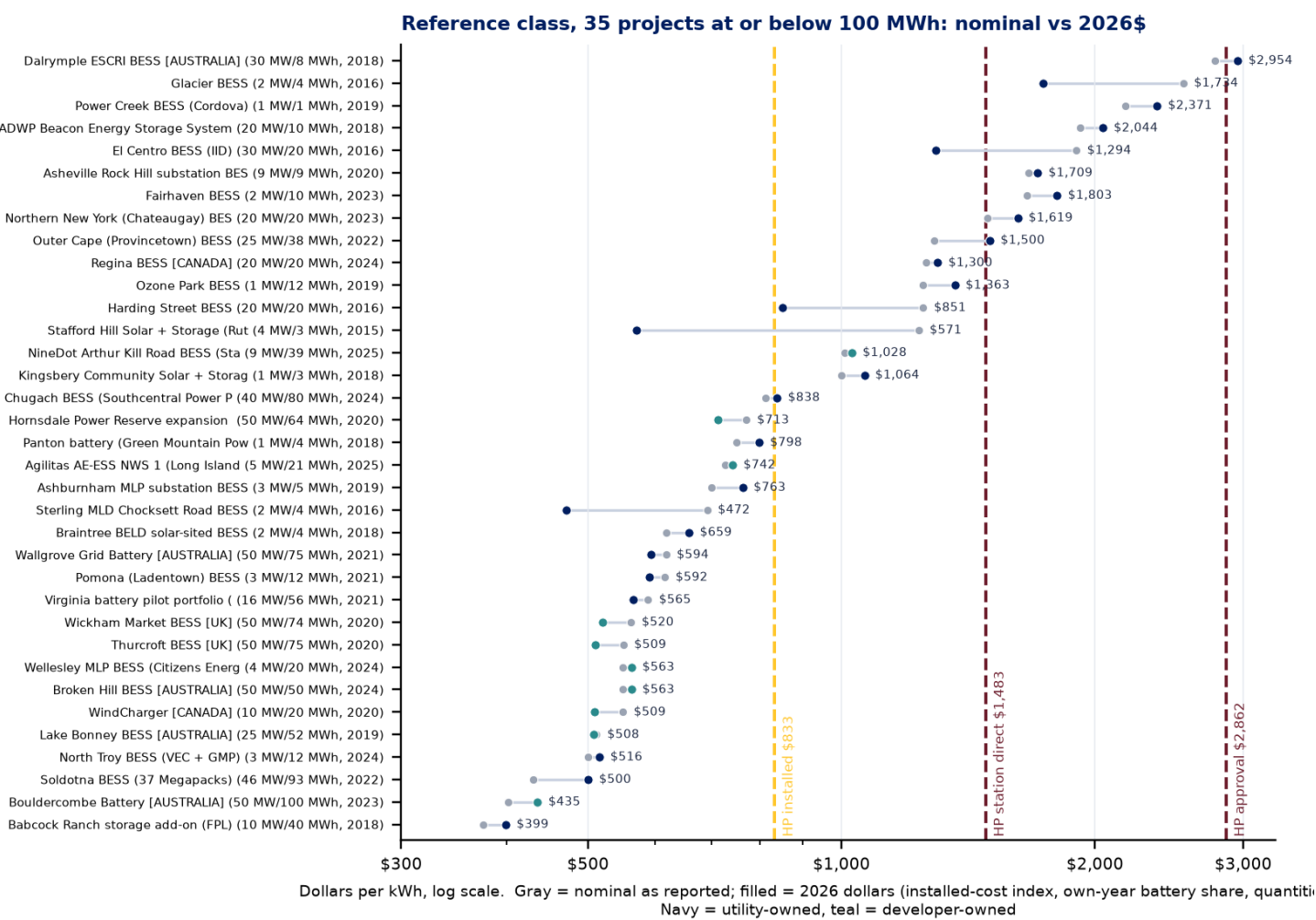
Item	\$	\$/kWh
Escalation in the approval level (Sta 360 \$2,622K + Sta 496 \$326K)	\$2,948K	\$147
on the Nomad balance, which Exhibit C fixes until Final Acceptance	\$1,130K	\$57
on station, site, feeder protection, testing and owner scope	\$1,818K	\$91
Spend curve it is applied to: actuals \$5.2M, 2026 \$13.5M, 2027 \$12.5M, 2028 \$26.1M		

Same comparison, same dollars (per kWh)

Boundary	Nominal	Ex-escalation	Benchmark brought to 2028 at 3%/yr
Vendor contract	\$605	\$605 (fixed)	BNEF US turnkey \$219 (2025) = \$239
Battery as carried (A)	\$650	\$593	NREL 4-h \$334 (2024\$) = \$376
Battery installed (A+B+C)	\$833	~\$756	NREL adjusted to 2-h, 10 MW: \$750-\$870
Station complete, direct	\$1,483	\$1,352	EIA/S&L \$436 (2023\$) = \$505; LBNL \$458 = \$516

- **Rates.** ESF template escalation, about 3% a year, on the 2026-2028 spend curve. \$2.9M is 5.1% of the approval level and 7.4% of directs. Benchmarks are 2023-2025 dollars; Rev 3 is 2026-2028 nominal, so one side must move before they are compared.
- **Decision pending.** \$1.13M sits on a price the contract fixes. Remove it or move it to the risk register with a stated rationale (open item, slide 37).
- **Brought to the same year, the installed battery is on the index.** NREL's \$334 is \$376 in 2028 dollars and \$750-\$870 after the 2-h and 10 MW adjustments; Hyde Park installed is \$833.
- **What 3% does not cover.** Lithium carbonate and tariffs moved faster than 3% in 2026. The battery is fixed by contract, so the exposure is the \$3.7M of non-battery material and the labor lines. The \$1.0M tariff item in the risk register is sized on total material, battery included.
- **Schedule.** Twelve months of slip adds about \$1.2M of escalation on the \$38.6M scheduled for 2027-2028, before AFUDC, PM and indirects (about \$5M in total).
- **Ex-escalation values** strip the station's \$131/kWh from the direct boundaries; the B+C share (~\$20/kWh) is allocated pro rata and is approximate.

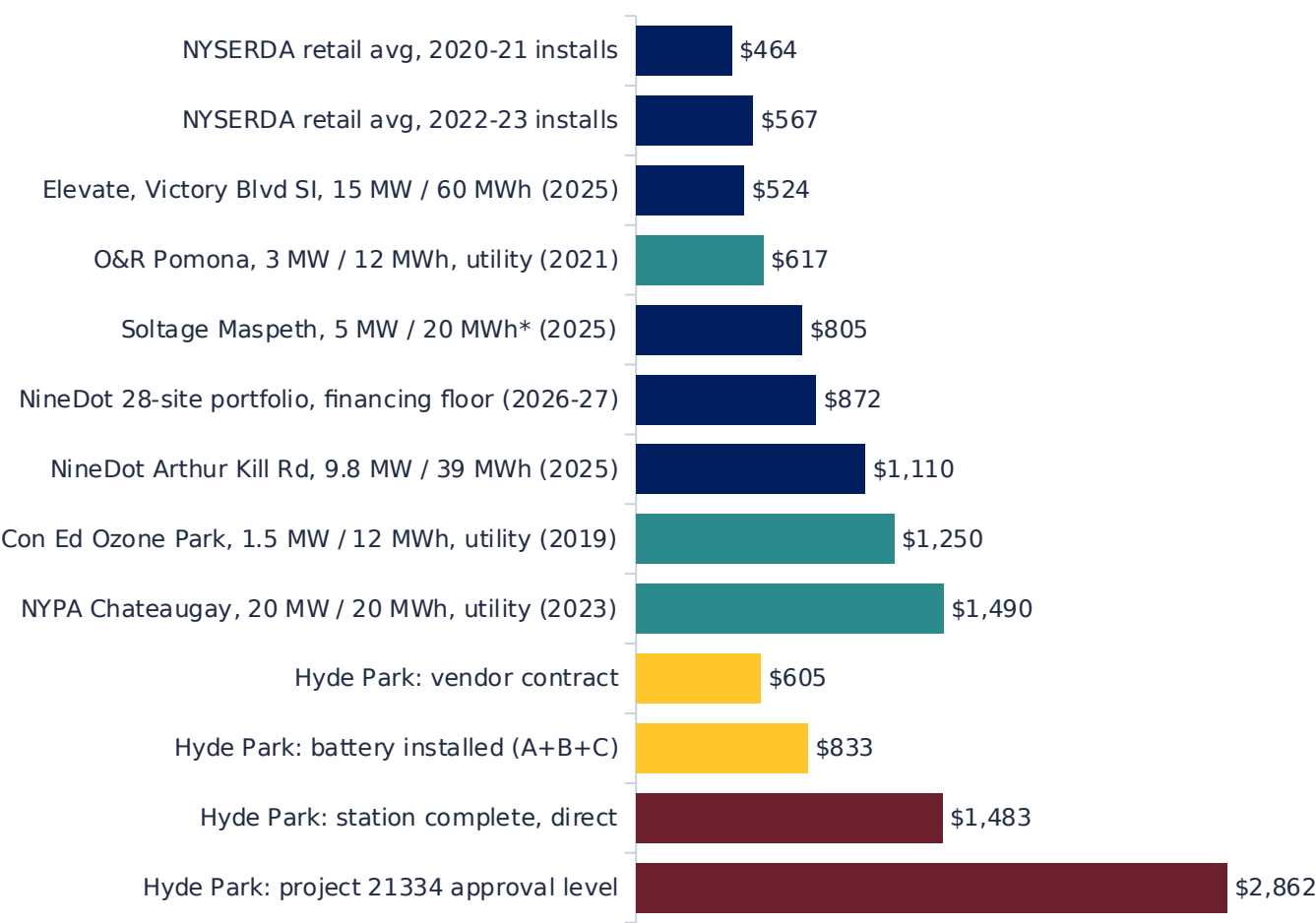
Own-year battery shares: the re-based class median barely moves and Hyde Park's station cost stays in the top fifth



Battery index (own-year share)	Median \$/kWh	HP installed	HP station
Nominal, as published	\$750	57%	77%
Installed all-in (EIA, LBNL)	\$742	57%	77%
BNEF pack price	\$743	60%	83%
BNEF turnkey system	\$686	57%	83%
NREL bottom-up base year	\$742	54%	80%
Lazard LCOS midpoint	\$892	40%	66%
Construction index only	\$1,052	40%	63%

- **The weights move with the prices.** Quantities are held fixed, not today's dollar shares: a 2016 project was about 73% battery and comes to 2026 at 0.68x; a 2018 project was 56% battery, 1.06x; 2021 and later about 1.0x. The 5 Oct review challenge stands: the earlier version held a 40% share fixed and made every old project dearer (median \$838).
- **Widening the class to 35 projects, 10 developer-owned,** puts Hyde Park's station-direct cost at the 77% percentile nominal and 77% re-based; the developer sites carry no utility station, no landfill and no siting case.
- **The battery curve you pick moves the median by 8%;** slides 14 to 18 show the curves and what moves it more.

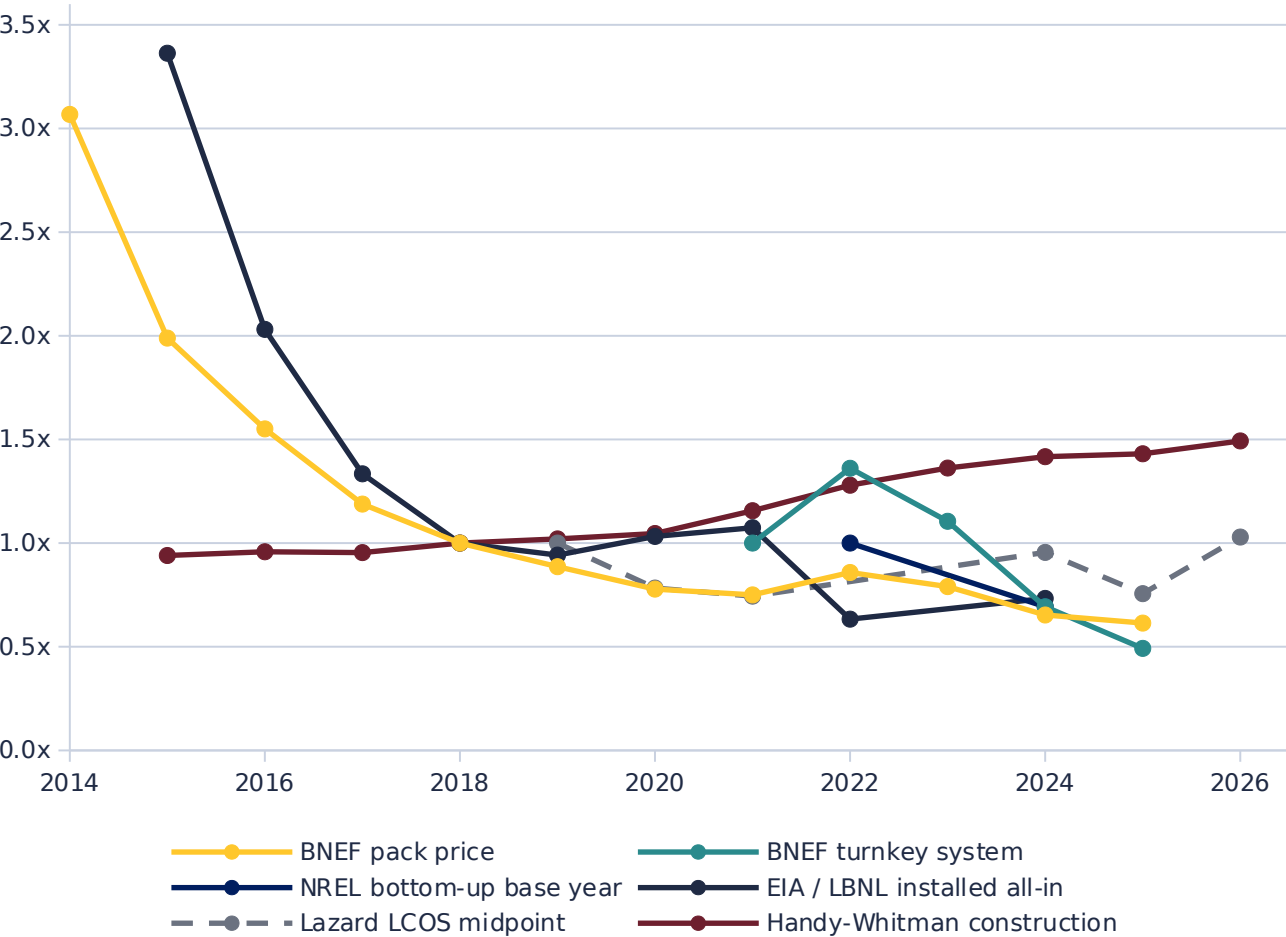
New York is building the same thing: distribution-connected, 2025, \$524 to \$1,110 per kWh before any utility station



Per kW of power	\$/kW
NineDot Arthur Kill Rd (NYCIDA budget)	\$4,427
NineDot portfolio financing (floor)	\$3,476
Soltage Maspeth (NYCIDA)	\$3,221
Elevate Victory Blvd (NYCIDA)	\$2,096
Hyde Park station complete, direct	\$2,967
Hyde Park approval level	\$5,725

- **NineDot:** 11 projects at 6 NYC sites, 39 MW operating (Sep 2026); 28 more financed at \$431M for 494 MWh. Thirteen NYC developer sites with NYCIDA-filed costs are in the dataset (NineDot, Convergent, Soltage, Agilitas, Greenbacker, Elevate): \$410 to \$1,110/kWh, median about \$830. Developer-owned 4-h sites on leased lots, VDER revenue, NYSEERDA incentive, ITC; no utility station, no landfill, no siting case.
- **Their own filed budget for Arthur Kill Road is \$1,110/kWh and \$4,427/kW**, inside Hyde Park's installed-to-station band on \$/kWh at twice the duration.
- **Interconnection is the cost no index carries:** Con Edison's two-part test added about \$21M of upgrades per project on 34 projects (161 MW), per the NY-BEST and NYSEIA filing of 11 Mar 2026. Hyde Park's Station 496 scope is \$7.3M.
- ***Soltage duration assumed 4-h;** incentives and ITC not netted from any row.

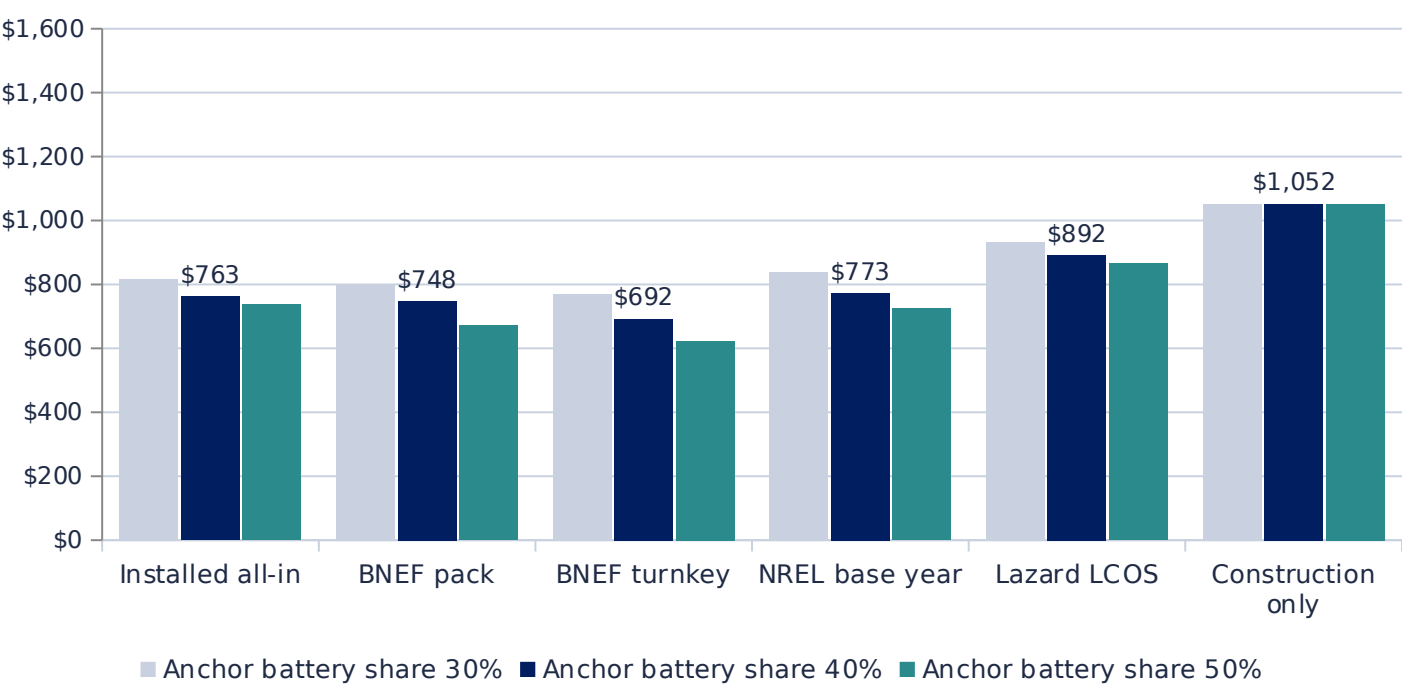
Six published curves, indexed to 2018: batteries fell 1.4 to 2x, the rest of a project rose half again



Curve	What it measures	2018 to latest	Use in re-basing
BNEF pack price	Cells and pack, ex-works; real \$; annual since 2010	0.61x	Battery share only
BNEF turnkey system	Racks, PCS, BOS, install; excl. grid connection, land, owner costs	0.49x	Battery share; short history
NREL bottom-up base year	Modeled 4-h 60 MW all-in incl. EPC, developer overhead, interconnection allowance	0.69x	Battery share; two points
EIA / LBNL installed all-in	Reported capitalized cost of installed systems (EIA-860 2015-18; LBNL 2024)	0.73x	Primary: closest to a project
Lazard LCOS	Levelized \$/MWh incl. charging, O&M, augmentation, financing, ITC	1.03x	Not a capital-cost deflator
Handy-Whitman construction	Utility plant construction cost index, North Atlantic (via PJM)	1.49x	Non-battery share (60%)

■ **Four curves describe the battery, one the rest of the project, one neither.** A re-basing picks one from each of the first two groups and says which. LCOS is cited most and fits least.

The battery curve you choose moves the median 8%; the battery share and the construction index move it more



Nominal median \$750. Hyde Park station complete, direct, is \$1,483/kWh; its percentile under each index is at right. Deflating whole projects at the battery rate gives \$390 to \$549.

Indices as published (BNEF pack and ESS surveys 2014-2025; EIA-860 2015-18 and LBNL 2024; NREL 2022 and 2024 base years; Lazard LCOS v5.0-2026 midpoints; Handy-Whitman via PJM), linearly interpolated between published points and held flat outside them. Retrieved from search excerpts; verify before external use.

Index	2018 to 2026	Factor to 2026: 2018 / 2016 project	HP station pct
Installed all-in	0.73x	1.06x / 0.68x	77%
BNEF pack	0.61x	0.97x / 0.74x	83%
BNEF turnkey	0.49x	0.92x / 0.93x	83%
NREL base year	0.69x	1.03x / 1.05x	80%
Lazard LCOS	1.03x	1.26x / 1.29x	66%
Construction only	1.49x	1.49x / 1.56x	63%

- **Choice of battery curve:** medians \$686 to \$743 across the four battery series (LCOS \$892). The curves disagree about packs, not about projects.
- **Anchor battery share:** 30% to 50% (2024 basis) moves the median \$79 to \$146 depending on the curve; the share is a lever of the same order as the curve.
- **Construction index:** Handy-Whitman North Atlantic rose 49% from 2018 to 2026 (10% a year in 2021-22); a flat 3.5% proxy gives 32%, and a median \$742 instead of \$742.
- **LCOS** is a levelized \$/MWh metric, flat since 2019; using it re-bases nothing on the battery side.

Whichever curve you cite, Hyde Park's station cost stays in the top quarter and the approval level stays above the class

Curve used to re-base	Class median	HP installed pct	HP station pct	Utility subclass median	HP station pct (utility)	Whole cost deflated at battery rate: median / HP station pct
Nominal, as published	\$750	57%	77%	\$1,235	68%	n/a
EIA / LBNL installed all-in	\$742	57%	77%	\$851	68%	\$549 / 86%
BNEF pack price	\$743	60%	83%	\$929	76%	\$517 / 94%
BNEF turnkey system	\$686	57%	83%	\$915	76%	\$390 / 100%
NREL bottom-up base year	\$742	54%	80%	\$1,189	72%	\$550 / 91%
Lazard LCOS midpoint	\$892	40%	66%	\$1,339	52%	\$875 / 74%
Construction index only	\$1,052	40%	63%	\$1,493	48%	n/a

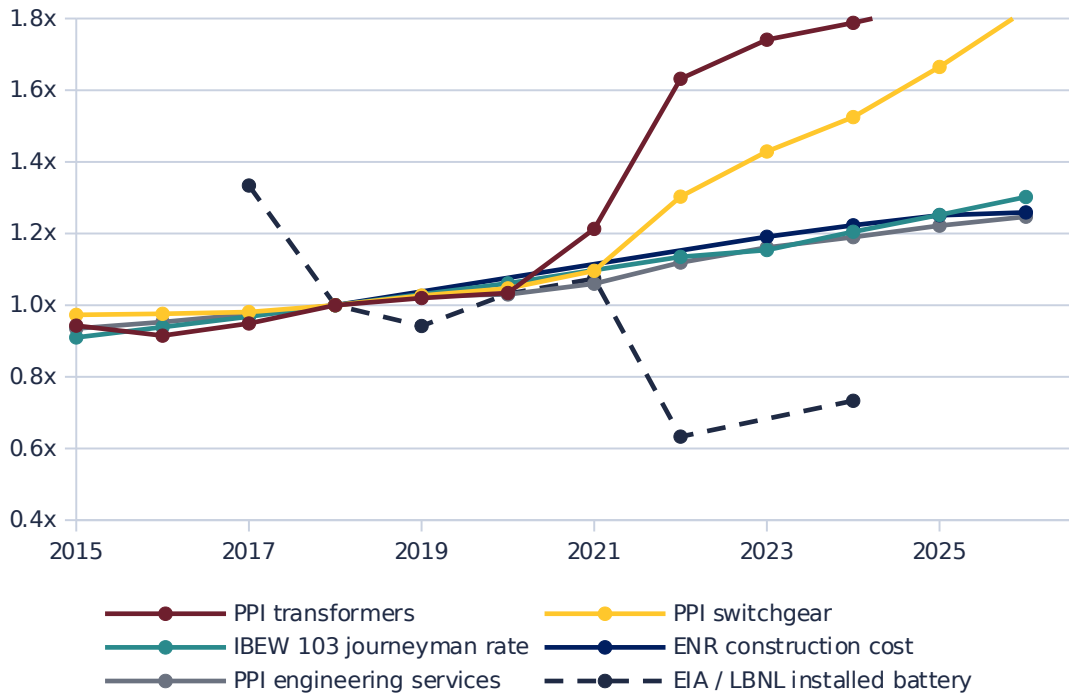
■ **Reading across:** the four battery curves place the class median within \$56/kWh of each other and Hyde Park's station cost between the 77% and 83% percentiles. LCOS and a construction-only index raise the class because they carry no battery deflation.

■ **The last column is the challenge as it was put to us:** deflate every comparable at the battery rate. It moves the median to \$390 to \$550 and Hyde Park to the top of the class, by treating switchgear, civil works and interconnection as lithium. It is the vendor framing applied backward in time.

■ **Which to cite:** the EIA / LBNL installed series to a regulator (reported costs of real installations); the BNEF pack series to a vendor (it is their number); neither LCOS nor a flat proxy to anyone for re-basing capital cost.

One escalator per layer, quantities fixed: a 2018 project like Hyde Park costs +4% in 2026 dollars, a 2016 one 32% less

Layer (loaded share)	Share	Escalator	2018 to 2026
A. Battery contract	38%	EIA / LBNL installed battery	0.73x
B. Battery install	5%	IBEW 103 journeyman rate	1.30x
C. GSUs, aux power	6%	PPI transformers	1.91x
D. Switching station	11%	PPI switchgear	1.82x
E. Site and civil	8%	ENR construction cost	1.26x
F. Feeder protection (496)	13%	IBEW 103 journeyman rate	1.30x
G. Test and commission	4%	IBEW 103 journeyman rate	1.30x
H. Owner soft costs	13%	PPI engineering services	1.25x
I. Taxes, insurance	2%	Handy-Whitman	1.49x
Composite, Hyde Park scope	100%	fixed quantities (1 / weighted sum of inverse ratios)	1.04x



Each series divided by its 2018 value. BLS PPI series are exact to 2023 (FRED mirror), excerpt anchors after; ENR and IBEW from published tables.

- **Five non-battery escalators sit between 1.25x and 1.91x since 2018; the battery sits at 0.73x.** With Hyde Park's physical scope held fixed, a 2018 build comes to 2026 at 1.04x (two-index model 1.06x) and a 2016 build at 0.68x, because the battery was 72% of such a project in 2016 and is 38% now. Transformers (1.91x) and switchgear (1.82x) are the fastest risers.
- **The review challenge was right about 2015 to 2017, and the earlier version of this slide was wrong.** Holding today's 38% battery share fixed made a 2016 project 1.08x; letting the share be what it was then makes it 0.68x. From 2018 on the battery decline and the construction rise roughly cancel. Deflating a whole project at the pack rate is still wrong: it gives 0.36x for 2016.

Three defensible methods, one answer: class median \$742 to \$746, Hyde Park's station cost at the 77% percentile

Method	What it does	Class median	HP installed pct	HP station pct	Utility median	HP station pct (utility)
As published	no adjustment	\$750	57%	77%	\$1,235	68%
Whole cost at the battery rate	vendor framing applied backward	\$549	69%	86%	\$685	80%
Two indices, own-year battery share	anchor 40% utility, 55% developer; slides 12 to 16	\$742	57%	77%	\$851	68%
One index per layer, developer weights	NREL/PNNL composition	\$744	60%	77%	\$854	68%
One index per layer, Hyde Park weights	Rev 3 loaded layers	\$746	57%	77%	\$853	68%
Layer method, BNEF pack for the battery	sensitivity	\$680	63%	83%	\$832	76%
Layer method, BNEF turnkey for the battery	sensitivity	\$585	63%	89%	\$770	84%
Earlier version: today's 40% share held fixed	superseded 5 Oct 2026	\$838	49%	71%	\$1,285	60%

How to scale any published BESS cost to 2026 (the workbook tab does it)

- **1. Fix the dollar year and the scope boundary.** Contract, installed, station or all-in; real or nominal; which year's survey.
- **2. Fix the battery share in today's prices, then back-cast it.** 40% for an all-in utility project, 55% for a developer site; the two indices tell you what that share was in the project's year (about three quarters in 2016).
- **3. Apply a battery index to the battery share and a construction or layer index to the rest, holding quantities fixed.** Never one index to the whole cost, and never today's shares on yesterday's prices.

- **Method spread:** \$742 to \$746 across the three defensible methods; \$549 for the naive one.
- **Hyde Park:** installed 57% to 60% percentile, station 77% to 77%, under every defensible method.

Six things about Hyde Park that a 100 MW greenfield benchmark never has to pay for

Size. 10 MW / 20 MWh. A switching station, a siting case and a commissioning team cost about the same at 10 MW as at 100 MW; per kWh they are ten times heavier.

Duration. Two hours. Fixed \$/kW items (PCS, breakers, GSUs, interconnection) are spread over half the MWh of a four-hour benchmark.

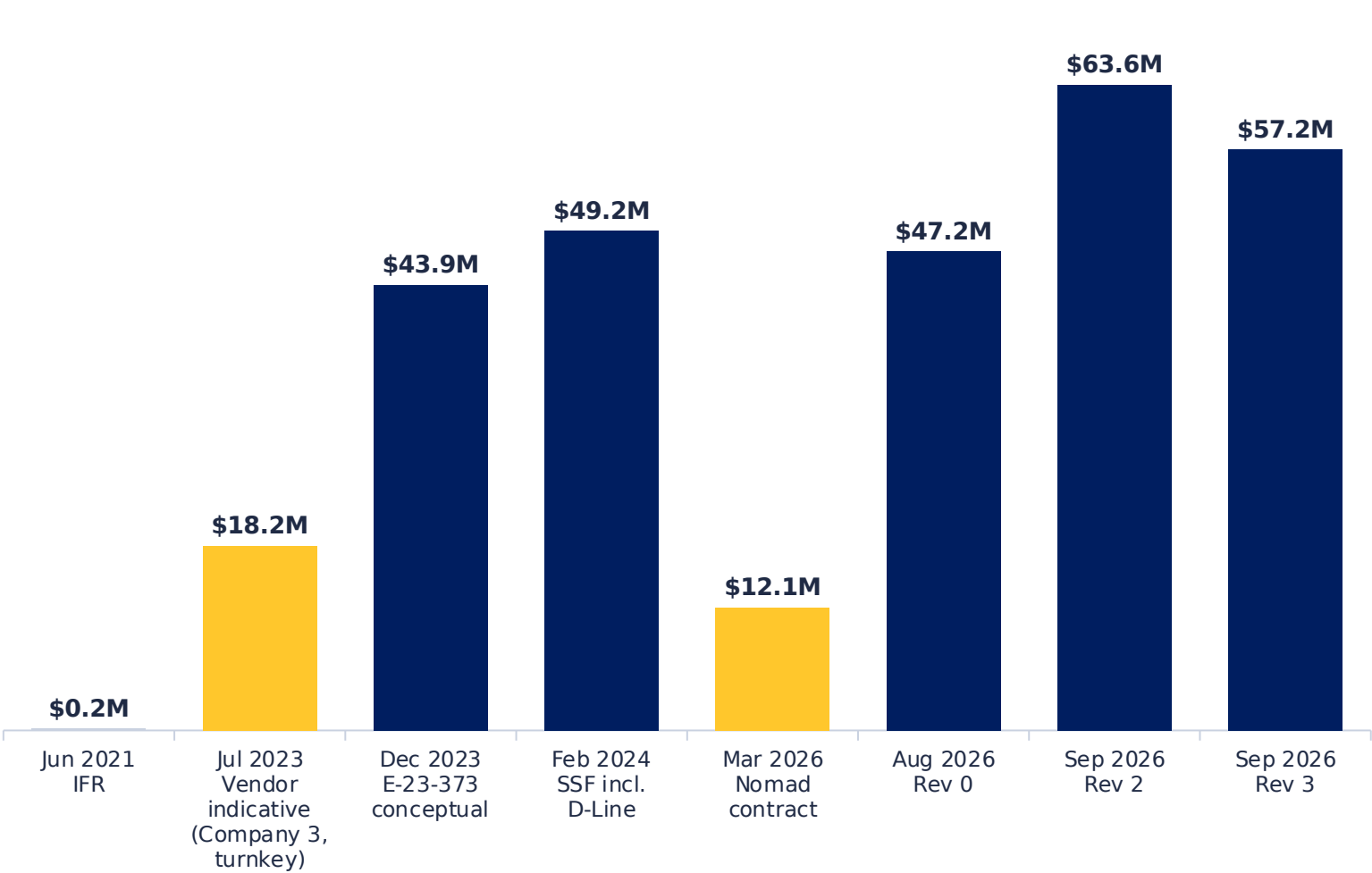
Site. A closed construction-debris landfill in Boston: DEP post-closure use permit, cap protection, 36 borings, imported platform fill, 14 ft retaining wall, unknown obstructions (\$450K risk).

Interconnection. Four feeder tie-ins at 13.8 kV, each with its own breaker skid, plus relay replacement and transfer trip at Station 496 (\$7.3M) and a 5,465 ft circuit extension (24211).

Siting and outreach. Article 80 Small Project Review, Zoning Board of Appeal, outside counsel, an EJ community, a neighborhood concerned about open land and views: \$1.5M plus \$0.5M in the risk register.

Utility accounting. Corporate indirects (\$10.0M) and AFUDC over a seven-year development window (\$5.0M) are real costs to customers that a developer's turnkey price never shows.

Five years, five in-service dates, and an estimate that moved from \$44M to \$57M

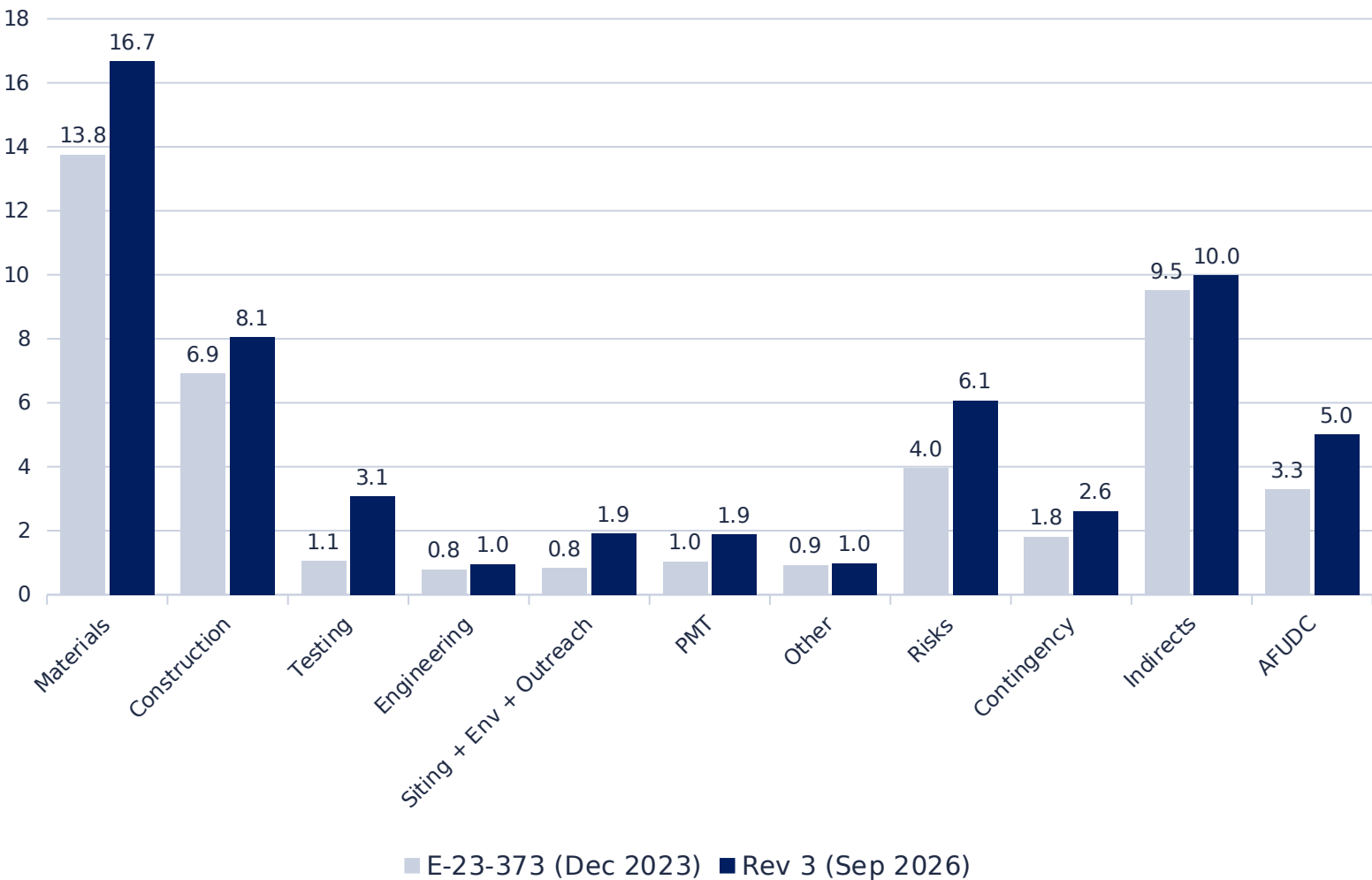


In-service date drift

As of	Planned ISD
Jun 2021	Q3 2025
Dec 2023	Apr 2027
Feb 2024	Sep 2027
Jan 2026	Aug 2028
Jun 2026	Dec 2028

Gold bars are vendor prices for the battery scope only. Navy bars are full project estimates. The battery price fell from \$18.2M turnkey-indicative (2023, incl. civil) to \$12.1M contracted (2026); the project rose \$13.3M in the same window. The 2023 conceptual carried a placeholder that was never re-based until August 2026.

What changed between the December 2023 conceptual and Rev 3: \$43.9M to \$57.2M

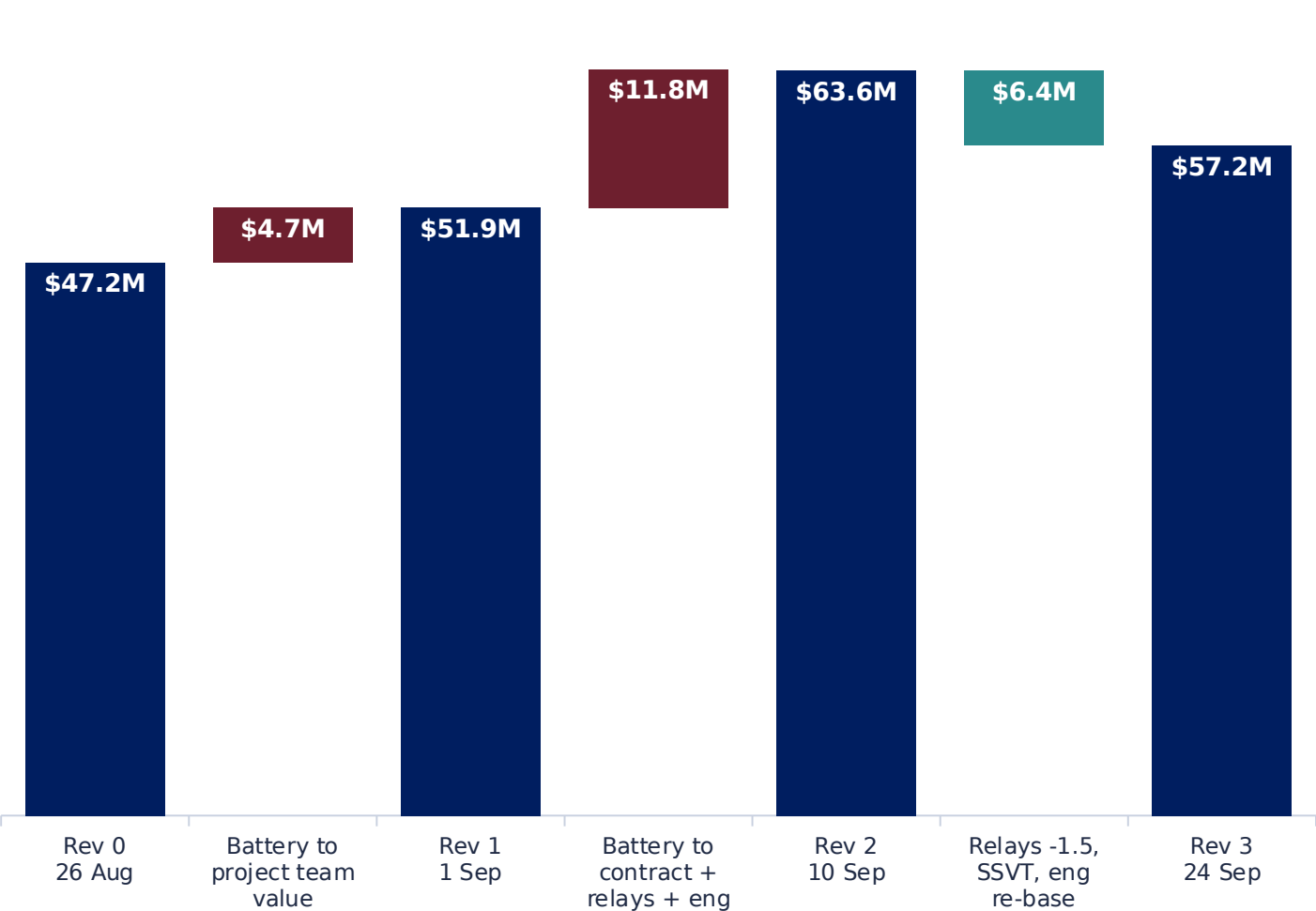


Largest movements

Category	Delta
Materials	+\$2.9M
Risks	+\$2.1M
Testing	+\$2.0M
AFUDC	+\$1.7M
Construction	+\$1.1M
Siting + Env + Outreach	+\$1.1M
PMT	+\$0.9M

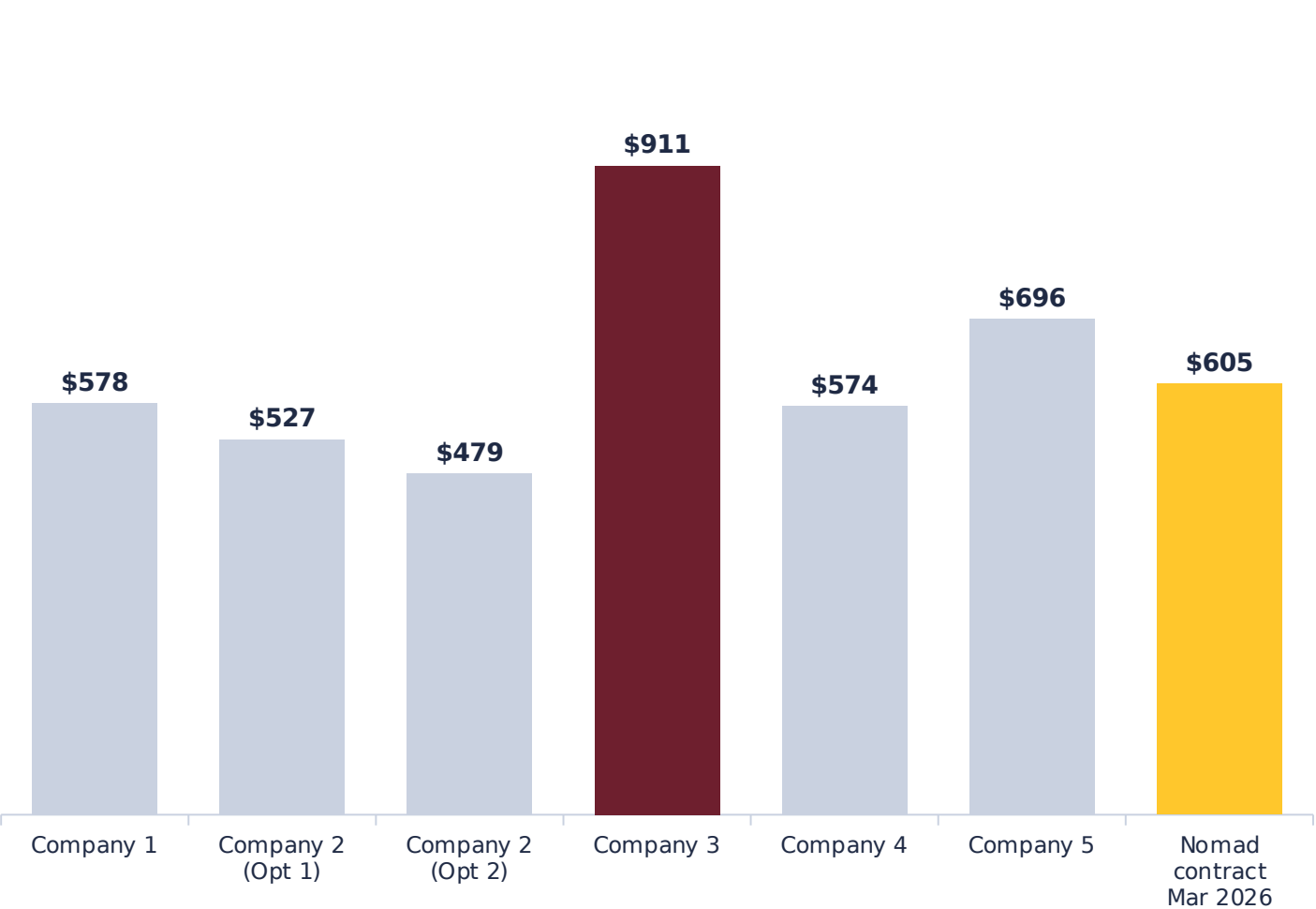
Materials: contracted battery replaces 2023 indicative pricing, plus breaker skids and GSUs. Testing and siting: Station 496 scope and Article 80 were not in E-23-373. AFUDC: a 20-month later ISD on a larger base.

Four revisions in five weeks: how the September estimate settled at \$57.2M



- **Rev 0 carried the battery at about \$6M** from the 2023 indicative sheet. The PM: 'we know the BESS system itself is 12.'
- **Rev 2 reset it to the cover agreement, escalated to \$13.1M**, and carried \$1.5M of 'relays for batteries' from a 2023 bidder that priced its own switchgear.
- **Rev 3 removed the relays** (P&C Engineering: included in the \$300K Siemens skid), cut three SSVTs to one, and re-based engineering to \$501K to go plus actuals.
- **Net:** +\$10.0M from Rev 0, of which \$7.0M is the battery line catching up to the contract.

The battery: a \$9.6M to \$18.2M spread in 2023, a \$12.1M fixed price in 2026



- **July 2023, five bidders, 10 MW / 20 MWh:** \$9.6M to \$18.2M. Every bidder excluded interconnection, permits beyond local, below-grade risk, sound walls or spares.
- **Company 3 (\$18.2M, burgundy)** became the E-23-373 basis because it alone priced turnkey civil (\$4.56M) and installation.
- **March 2026 contract, \$12.1M (gold):** supply only, with supervision. Civil, install, GSUs, station and interconnection moved to Eversource scope.
- **Equipment-only inside the contract:** \$10.64M, \$532/kWh, in line with published 2025-26 turnkey system indices (slides 9-10).
- **Rev 3 adds \$1.13M of escalation** to a price the contract fixes until Final Acceptance.

Three assumptions that, if wrong, move the number more than the reserve

1. Construction may be low, not high.

Rev 3 layers B+D+E are \$8.3M direct. Haugland's Class 3 base for overlapping scope is \$19.2M with a \$2.9M deduct ceiling. Nobody has mapped one to the other.

Settles it: a signed HEG-to-Rev 3 scope map before the PAF is drafted.

2. Siting is binary and carries no float.

ZBA determination, construction start and container delivery all land in Q1 2028 for a Dec 2028 ISD. Denial means an EFSB exemption at 18-24 months. The estimate carries \$250K for siting delay.

Settles it: Article 80 filing date, ZBA calendar, P6 float on the siting chain.

3. The number is Class 4 wearing Class 3 clothes.

30% engineering, unit-library construction with no bid, install labor as a percentage of a 2023 total, a 2.5 ft platform placeholder. EPAC will hear \$57.2M as a planning number.

Settles it: geotech, HMA, HEG map, crew build-up; then reclassify with a -15/+25% range.

Stress tests on our own numbers, and what history says about small utility batteries

Scenario	Loaded cost	Date	Covered by \$7.6M reserve?
Construction unit costs +25% on B+D+E+G	+\$4.0M	0	Yes, alone
HEG reconciliation at midpoint of the gap (+ \$5M direct)	+\$8.4M	0	No
Schedule +12 months (siting or long-lead)	~+\$5M	+12 mo	Partly
EFSB zoning exemption path	~+\$8M to + \$10M	+18-24 mo	No
Contractor bids +20% over estimate labor	+\$2.1M	0	Yes, alone
Combination: HEG midpoint + 12-month slip	~+\$13M (to ~\$70M)	+12 mo	No
Battery change orders at the LD cap (10%)	+\$1.2M	0	Yes

Untestable from the package: tariff or change-in-law pass-through on the Nomad price (Exhibit C terms), the AFUDC rate (assumed ~7% on CWIP), Station 496 outage premium (1.35 to 1.70 factor on affected hours), FEOC / ITC effect on net cost.

Base rate: approved to actual

Project	Uplift
BGE Fairhaven 2.5 MW/9.7 MWh	+64%
HECO Waena 40 MW/160 MWh	+37%
SaskPower Regina 20 MW	+31%
Eversource Provincetown	+9% to +40%
Hyde Park, Dec 2023 to Sep 2026	+30%
Hyde Park, Rev 0 to Rev 3 (5 wks)	+21%

Rev 3 carries 17.9% risk plus contingency. The exception case for beating the base rate rests on three anchors: a fixed battery price, a Siemens price on the skids, and Haugland engaged pre-construction. Together they cover about a third of the cost. The other two thirds have no such anchor.

Pre-mortem: June 2031, \$74M, fifteen months late. What happened?

The ZBA conditioned approval on a smaller footprint and new screening; the civil package went back through Haugland in the winter of 2028. Spring 2027 borings found debris deeper than the 2014 closure set showed; the platform went from 2.5 ft of fill to piles, and the reconciliation that had never been done was done as a change order. Nomad's twelve containers arrived on time in January 2028 and sat in a rented yard for nine months. Station 496 cutovers needed night outages priced at straight time. AFUDC alone added \$4M. The PAF had said Conceptual, -25/+50%. Everyone remembered \$57M.

Early-warning indicators

Indicator	Caught today?
Geotech report not issued by IFC-civil design freeze (mid-2027)	No: no P6 gate ties IFC to it
ZBA hearing date not set within 90 days of Article 80 filing	No: siting carries hours, not milestones
HEG reconciliation not closed before the PAF is drafted (Nov 2026)	Only if the CoE makes it a precondition
Nomad final-drawings milestone (13 Jul 2026) slipped	Partly: payments tracked, not downstream effect on GSU spec and civil
Station 496 outage plan not issued before construction pricing	No

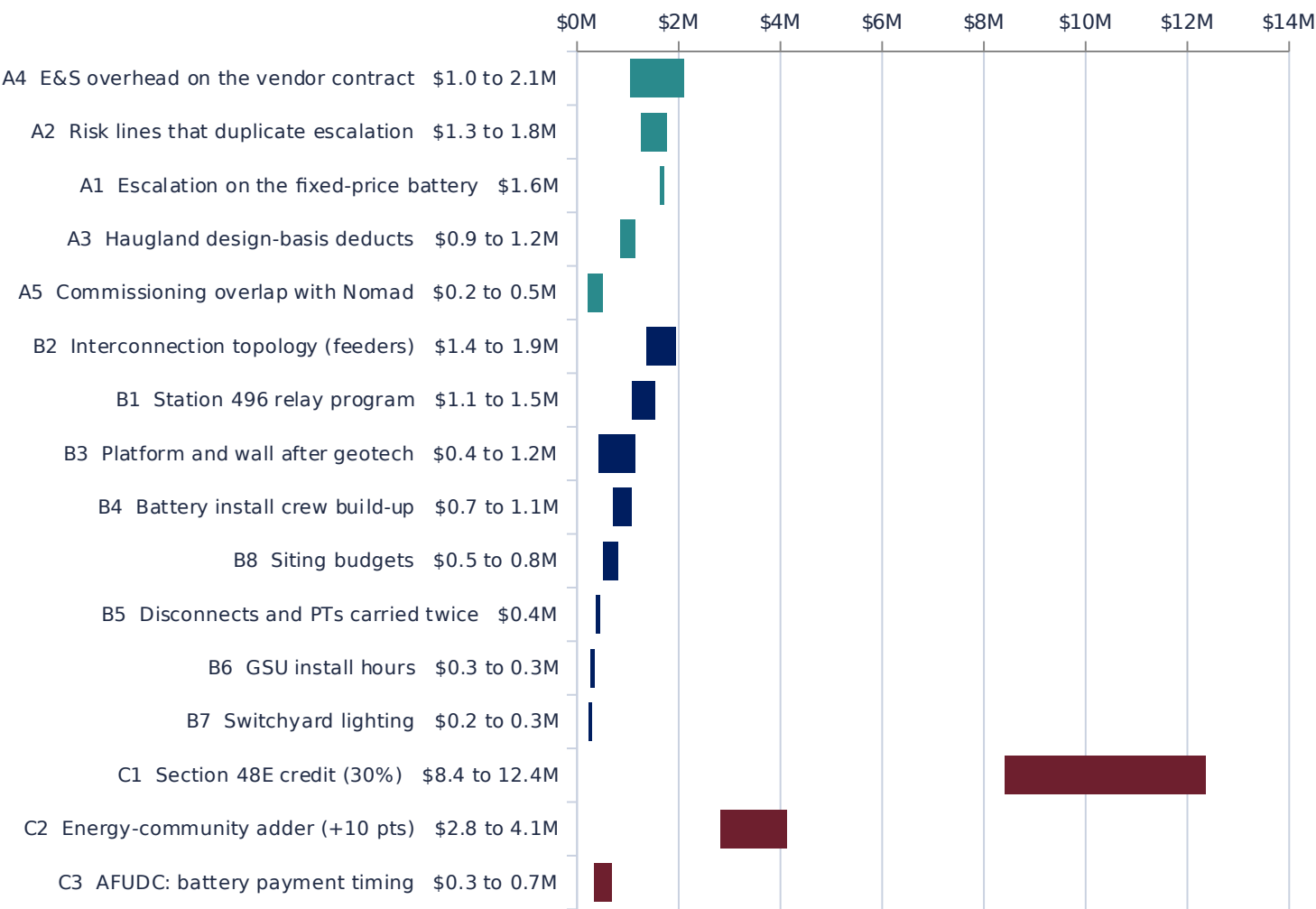
16 levers in three tiers: \$5.0 to 7.1M in the estimate, \$4.9 to 7.5M in scope decisions, and a tax credit larger than both

#	Lever	Layer	Direct	Loaded*	Conf.	What must close
A1	Remove escalation carried on the fixed-price Nomad balance	A	\$1.1M	\$1.6M	High	Exhibit C read for tariff and change-in-law pass-through; decision minuted
A2	Risk register: labor-cost and material-cost lines overlap the escalation already in the directs	J	\$1.3 to 1.8M	\$1.3 to 1.8M	Medium	Re-word the two lines as labor-shortage and tariff risks and size them on exposure, not on the whole material and labor base
A3	Confirm the four written design-basis items Haugland priced as deducts	D/E	\$0.6 to 0.8M	\$0.9 to 1.2M	Medium	Retaining wall, basin, duct bank and GSU count confirmed in writing; HEG map signed first
A4	Engineering & supervision overhead basis applied to the vendor turnkey contract	L	\$1.0 to 2.1M	\$1.0 to 2.1M	Low	Ruling on whether the E&S station basis applies to vendor-furnished, vendor-commissioned equipment
A5	Commissioning hours carried from E-23-373 overlap Nomad's commissioning support	G	\$0.1 to 0.3M	\$0.2 to 0.5M	Low	Commissioning split (Nomad vs owner) written into the test plan
B1	Station 496 relay program: 710 man-hours per relay	F	\$1.1 to 1.5M	\$1.1 to 1.5M	Medium	Two recent relay-replacement actuals; confirm the four SEL-735 are protection, not metering; outage windows priced
B2	Interconnection topology: five feeder cabinets, four GSUs, four breakers, four metering sets for 10 MW	C/D	\$0.9 to 1.4M	\$1.4 to 1.9M	Low	Hosting-capacity study: can two feeders take 5 MW each? If not, the item is zero
B3	Platform and retaining wall sized after geotech, not before	E	\$0.3 to 0.8M	\$0.4 to 1.2M	Low	Geotech report; MassDEP SW45 post-closure permit conditions; NFPA 855 spacing
B4	Battery installation labor from a crew build-up instead of 25% of a 2023 total; fire suppression bought twice?	B	\$0.5 to 0.7M	\$0.7 to 1.1M	Medium	Crew build-up: crane set, anchoring, DC and AC terminations per Voyager unit; confirm whether the Voyager enclosure carries its own suppression (\$132K carried)
B5	Disconnect switches and PT cutouts priced as separate lines although the Siemens skids carry them	D	\$0.3M	\$0.4M	High	Strike lines 07.03.02.04, 06.02.09.00 and the PT cutouts once Siemens confirms the skid scope in writing
B6	GSU installation hours: 524 man-hours per pad-mount; count to be settled (four carried, twelve mentioned)	C	\$0.2 to 0.2M	\$0.3 to 0.3M	Medium	Topology and GSU count in writing; crew build-up for a pad-mount set; 2026 transformer quote with the 29-70 week lead time in the schedule
B7	Switchyard lighting: 12 HPS luminaires and \$200 a foot of hookup on an unmanned one-acre pad	D	\$0.2M	\$0.2 to 0.3M	Medium	Lighting design to the security need (fence-line LED, motion); hookup length and rate from the layout
B8	Siting and project management: external siting at \$1.05M and a \$157K siting line inside Station 496	H/F	\$0.4 to 0.6M	\$0.5 to 0.8M	Low	Task budget from outside counsel for Article 80 small-project review and the ZBA permit; confirm what the Station 496 siting application is for
C1	Section 48E investment tax credit on the storage property	rev. req.	\$8.4 to 12.4M	\$8.4 to 12.4M	Medium	Eligible-basis opinion; FEOC cost-ratio test (65% non-PFE for a 2028 construction start); normalization election; DPU treatment
C2	Energy-community adder if the closed landfill qualifies as a brownfield	rev. req.	\$2.8 to 4.1M	\$2.8 to 4.1M	Low	Phase II ESA or a prior brownfield-program assessment (the Phase I safe harbor stops at 5 MW)
C3	AFUDC: pay the battery at delivery, not a year ahead; protect the siting float	M	\$0.3 to 0.7M	\$0.3 to 0.7M	Medium	Milestone payment schedule in the Nomad SOV aligned to 2028 delivery; float target on the ZBA chain

*Loaded at the marginal loader of 1.44 (contingency, basis-driven indirects and AFUDC scale with directs; the risk register and booked actuals do not), not the 1.685 average used on slides 5 and 25. Tier C items reduce the revenue requirement, not the \$57.2M. Tiers A and B are gross of the Haugland reconciliation (red team KS1, +\$4M to +\$8M direct), which must close first.

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What the levers are worth, loaded, and how much of it survives the red team's reconciliation



Tier	Loaded range	Against
A. Estimate hygiene	\$5.0 to 7.1M	the \$57.2M
B. Value engineering	\$4.9 to 7.5M	the \$57.2M
A + B	\$10.0 to 14.6M	the \$57.2M
Haugland gap (KS1)	+\$4 to 8M direct	the \$57.2M
C. ITC, adder, AFUDC	\$11.6 to 17.2M	revenue requirement

■ **Net of the red team:** Tiers A and B (\$10.0 to 14.6M) and the Haugland gap (+\$4 to 8M) are the same order of magnitude. The honest planning case is a \$57M number that holds, not one that falls.

■ **The three to pursue now, in order:** A1 and A2 (a decision and a re-wording, about \$3M), B1 (two phone calls to EMA for relay actuals), and C1 (a memo to Tax). Everything else waits on geotech, the hosting-capacity study or the HEG map.

■ **What not to do:** take Tier B into the PAF before the studies close. A VE deduct booked ahead of its study is how estimates go to \$74M.

The investment tax credit is worth \$8.4 to 12.4M to customers and nobody on the project owns it yet

\$8.4M to \$12.4M

30% credit, eligible basis

Basis \$28M (battery, install, BOP, loaded) to \$41M (plus station, site, testing).

+\$2.8M to \$4.1M

Energy-community adder

Only if the closed landfill qualifies as a brownfield. Not assumed.

65%

Non-PFE cost ratio required

Storage is exempt from the 2027 wind and solar cliff. For a 2028 construction start, 65% of direct cost must come from outside prohibited foreign entities.

What must close	Owner	Why it matters
Eligible-basis opinion: which of layers A to G are energy storage property; interconnection property (Station 496) is outside the storage definition in 48E	Tax	Sets the \$28M to \$41M range
FEOC cost-ratio test (Notice 2026-15): cells are 52% of equipment cost in the IRS table, so Chinese cells alone can fail a 65% threshold	Tax + Sourcing + Nomad	A failed test is a zero credit
Normalization election for storage (IRA amendment to section 50(d)(2)) and the DPU's treatment of the credit; no Massachusetts precedent yet (Outer Cape and Nantucket pre-date storage eligibility)	Regulatory	Decides how fast customers see it
Prevailing wage and apprenticeship written into the construction contracts	Sourcing + Construction	6% vs 30%
Brownfield status: a closed municipal landfill is not excluded, but above 5 MW the Phase I safe harbor is unavailable; needs a Phase II ESA or a prior program assessment	Environmental + Tax	The 10-point adder

- **Not an estimate change.** The \$57.2M stays; the credit reduces the revenue requirement the customer pays over the asset's life. It belongs in the PAF narrative and the rate case, not in the ESF.
- **Scale:** the low end of the credit is larger than Tiers A and B combined, and larger than the Haugland gap.
- **Risk:** the FEOC rules are the single question that can take it to zero. Ask Nomad for the cell and module origin now, before the final-drawings milestone.
- **Confidence:** medium on the mechanism (law-firm summaries of the July 2025 law and Notice 2026-15; statute not read first-hand), low on the dollars until Tax opines.

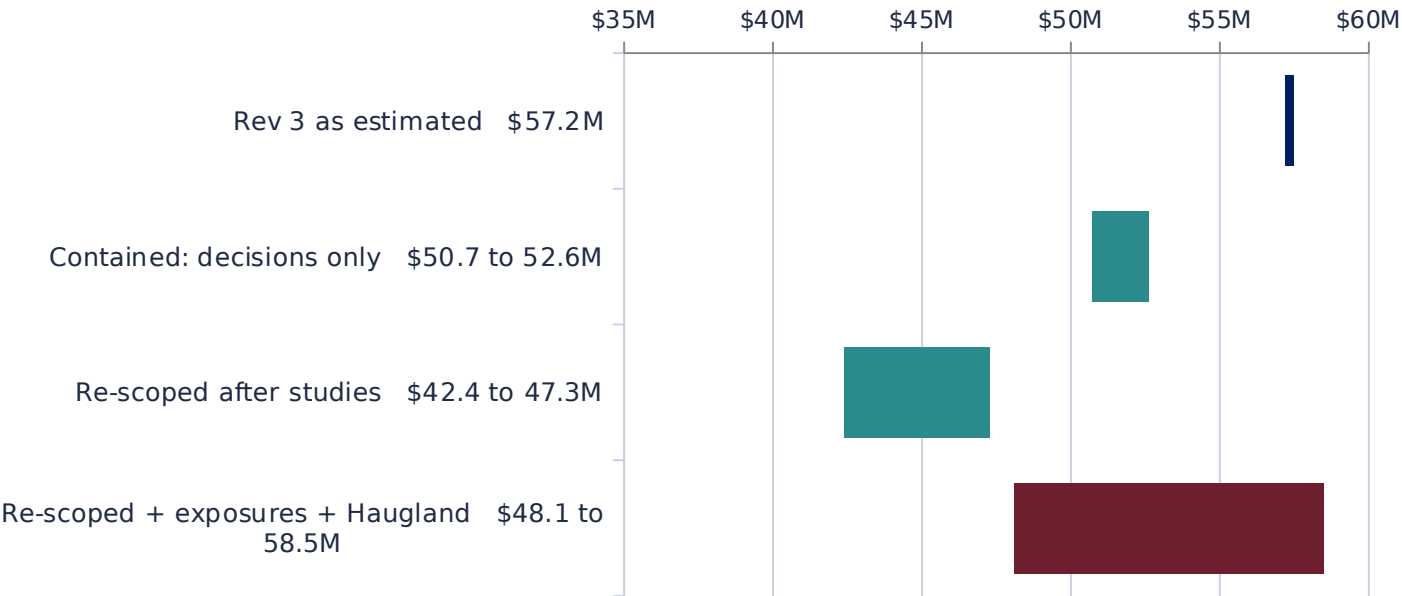
Scope challenge, layers A to E: a double count on the skids, a GSU count nobody has written down, and civil rates with no basis

Layer	Rev 3 carries	The function requires	Challenge	Direct \$
A. Battery contract	Nomad fixed price \$12.1M (12 Voyager units, 10-yr warranty \$0.80M, spares, support, shipping, training) plus \$1.13M escalation on the balance	The battery. Price is fixed until Final Acceptance.	Remove the escalation; align Attachment C-2 milestones to the January 2028 delivery; write the tariff and change-in-law clause into the change-order process; treat warranty years 3-10 as a service agreement if Accounting allows	\$1.1M
B. Battery installation	5,557 mh of install labor (25% of a 2023 site total), 12 pads, fire suppression \$132K, GSU-to-battery conduit and cable	Set, anchor and terminate 12 trailer-mounted units; Nomad supervises (two people, one day per container).	Crew build-up at 200-250 mh per unit (2,400-3,000 mh); confirm whether the Voyager enclosure carries its own suppression before buying \$132K of it twice	\$0.5 to 0.7M
C. GSUs and auxiliaries	4 x 3 MVA pad-mount GSUs \$1,077K furnish (2023 bidder list + 3%/yr) and \$363K install (2,098 mh = 524 per unit); comms \$355K; AC/DC panels \$137K	Step 480 V to 13.8 kV for 10 MW. The count follows the topology: one GSU per feeder connection or one per unit.	Settle the count in writing. The PM's 9 Sep email speaks of twelve GSUs; Rev 3 carries four; Haugland priced the larger count (\$632K deduct item). Install at 524 mh per pad-mount is three to four times a normal set; re-quote the transformer with the 2026 lead time (29-70 weeks) in the schedule	\$0.2 to 0.2M
C. GSUs (exposure)	Four GSUs	If the topology needs one GSU per Voyager unit, twelve	Eight more 1 MVA-class pad-mounts, foundations and cable	exposure \$0.6 to 1.0M
D. Switching station	4 Siemens breaker skids \$1,327K furnish (Siemens +/- 30% budgetary at \$300K each incl. relays, cabinets, DC battery, PTs, disconnects); 8 disconnect switches \$238K (1,040 mh); 4 metering transducers \$265K; lighting \$313K (1,272 mh); duct bank, 15 kV cable, riser poles, SSVT, grounding	Four feeder connections at 2.5 MW each because a 15 kV-class feeder takes about 9 MVA aggregate (National Grid MA limit; Eversource's not retrieved). Each connection needs a breaker, protection, metering and a tie.	(1) The 8 disconnects and the PT cutouts are on the skids per the project team and are priced again as separate lines: \$259K and 1,040 mh. (2) Lighting: 12 HPS luminaires and \$200 a foot of direct-bury hookup on an unmanned one-acre pad; LED on the fence line at a third of the cost. (3) Hosting-capacity study: if two feeders can take 5 MW each, two skids, two GSUs and two metering sets go away (savings register B2).	\$0.4M
E. Site and civil	Cut and re-place 2,520 CY \$456K (2,520 mh); 14-ft retaining wall 280 lf \$470K; trap rock 4,900 sy \$492K; gravel drive 760 lf \$319K; 36 borings \$361K; mob/demob \$313K	A stable, drained, accessible pad on a capped construction-debris landfill under a MassDEP post-closure permit; the wall, the platform depth and the cut volume are design outcomes of the geotech, not inputs.	Unit rates run two to four times library norms (cut at \$181/CY, surfacing \$100/sy, drive \$420/lf, borings \$10K each) on placeholder quantities (2.5-ft platform). No basis is written for either. The reconciliation with Haugland decides which way this layer moves; do not book a deduct before the geotech.	\$0.3 to 0.8M
E. Site (exposures)	No sound mitigation, no visual screening or faux facade, no police details beyond \$12K of traffic control, no NFPA 855 hazard mitigation analysis line	Constructability review: visual screening 'likely required' and a sound study to come; Boston public-way work; NFPA 855 for a lithium system next to housing	E-23-373 carried \$467K of sound mitigation; Rev 3 dropped it and the risk register does not carry it. Screening and a facade were deferred to 'vendor design'. Price them or register them.	exposure \$0.6 to 1.2M

Scope challenge, layers F to M: the relay program, the hours carried from 2023, the siting budget and the loaders

Layer	Rev 3 carries	The function requires	Challenge	Direct \$
F. Station 496 feeder protection	20 relays: hardware \$91K; install 6,400 mh \$1.23M; test 7,800 mh \$1.81M; E&S basis \$1.46M; siting application \$157K; AFUDC, contingency, risk	Directional protection on the four feeders that will see reverse flow, and coordination at the source station. Whether all 20 relays, the four SEL-735 meters and a siting application are needed for relay work inside an existing fence is the question.	\$363K and 710 man-hours per relay. Benchmark against two recent EMA relay-replacement actuals; confirm the SEL-735 are protection; confirm what the \$157K siting line is for; price the outage windows explicitly instead of inside the hours.	\$1.2 to 1.7M
G. Testing and commissioning	Field test 3,200 mh, LCE 1,600 mh, SCADA 1,650 mh, telecom 120 mh: \$1.26M, hours carried from E-23-373	Owner witness and acceptance testing; Nomad commissions its units (contract: \$210K support, \$246K vendor labor)	Write the split into the test plan; the owner's hours were set in 2023 before the vendor's scope existed	\$0.1 to 0.3M
H. Owner soft costs	Siting external \$1,052K plus internal and actuals; PM (contractor and Eversource, three lines plus actuals) \$1.30M; CM \$356K; engineering \$501K contractor plus actuals; outreach, environmental, legal	Article 80 Small Project Review and a ZBA conditional use permit; a PM and a CM for the construction window; 30%-to-IFC engineering	Siting at \$1,052K external is two to three times a typical Article 80 small-project budget; ask counsel for the task budget. PM hours are duration-driven: every month of siting slip adds about \$0.4M across PM, AFUDC and indirects, so the float target on the ZBA chain is the lever, not the rate.	\$0.2 to 0.4M
J. Risk register	\$5.3M Sta 360 + \$0.77M Sta 496: material \$1.0M, labor \$0.8M, design changes \$0.8M, UG obstructions \$0.45M, environmental, P&C and station engineering \$0.3M each, soil, weather, outreach, siting, BAR deductibles	Risks that the directs and contingency do not already fund	Labor-cost and material-cost lines restate escalation (savings register A2); design-change lines overlap the 17.9% contingency; sound mitigation and screening are the risks that are missing	\$1.3 to 1.8M
L. Indirects	\$8.06M: E&S station basis \$4.79M (16% of directs, applied to the vendor contract), general services, MDEC, vehicles, non-productive, AS&E, payroll	Template basis rates	Whether the E&S basis should load a vendor-furnished, vendor-commissioned contract is a policy question (savings register A4, up to \$2.1M); it is the single largest line nobody on the project can change	\$1.0 to 2.1M
M. AFUDC	\$5.0M over a seven-year development window	Capitalized carrying cost on CWIP	Pay the battery at delivery, not a year ahead (C3); protect the siting float; every month costs about \$0.4M all-in	\$0.3 to 0.7M

Three numbers: \$57.2M estimated, \$50.7 to 52.6M contained, \$42.4 to 47.3M re-scoped; honest range \$48.1 to 58.5M



- **Read it as a range of outcomes, not a menu.** The contained number needs only decisions and is worth taking to the November PAF. The re-scoped number is a ceiling on reductions; nobody should present it until the geotech, the hosting-capacity study and the Haugland map have closed.
- **What this says about the last-BESS question.** Even fully re-scoped, the station, the platform and the siting case stay in the number: they are the price of this site, not of batteries. A 10 MW battery inside an existing station fence on a standard design would carry neither. If the company builds another, that is the specification.
- **The ITC sits outside all four bars.** \$8.4 to 12.4M of revenue requirement, larger than any case's spread, and the FEOC answer from Nomad decides it.

Component	Loaded \$M	Needs
Reductions in the register	\$8.5 to 12.9M	decisions (A, J, L, M) and studies (B to H)
Two-feeder topology (B2), counted once	\$1.4 to 1.9M	hosting-capacity study
Exposures not carried	\$1.7 to 3.2M	sound, screening, details, NFPA 855, GSU count
Haugland reconciliation	+\$4 to 8M direct	signed scope map

Containment plan: five gates, a station-direct design-to-cost target of \$1,250 per kWh, and a monthly number someone owns

When	What	Who	What it decides
Now to Nov 2026 PAF	Decide A1 (battery escalation), re-word A2, get the Accounting ruling on the E&S basis, strike the disconnect and PT lines that the skids already carry, price or register sound mitigation and screening, open the 48E memo with Tax	Cost Estimating, PM, Accounting, Tax	Estimate hygiene: about \$3M to \$5M loaded; the exposures added honestly
Gate 1, design basis (Q1 2027)	Geotech report, NFPA 855 hazard mitigation analysis, hosting-capacity study for two feeders, GSU count and topology in writing, Haugland-to-Rev 3 scope map signed	Substation Eng., Civil, System Planning, HEG	Decides the two-feeder case (\$1.4M to \$1.9M), the civil layer (either way) and the GSU exposure
Gate 2, design-to-cost target (before IFC, mid-2027)	Set a station-direct target of \$1,250 per kWh (\$25M direct, the reference-class 75th percentile) and a loaded target of \$50M; monthly trend review against it; change control board with the CoE; no scope added without a funded source	PM (accountable), Cost Estimating (trend), sponsor	Turns the estimate into a managed number instead of a forecast
Gate 3, commercial strategy (before construction pricing, Q3 2027)	Competitive lump-sum pricing for station, civil and install (Haugland plus two), owner-furnished GSUs and skids on alliance pricing, Nomad milestones aligned to delivery, change-order cap, outage windows at 496 priced explicitly	Sourcing, PM, Construction	Converts unit-library construction into a bid; the red team's +25% construction stress test becomes a contract term
Standing	Monthly cost-containment report to the sponsor: station-direct \$/kWh against target, forecast at completion, open scope questions, exposures register; the 48E position and the FEOC cell-origin answer tracked as milestones	PM, CoE	What leadership sees every month until in-service

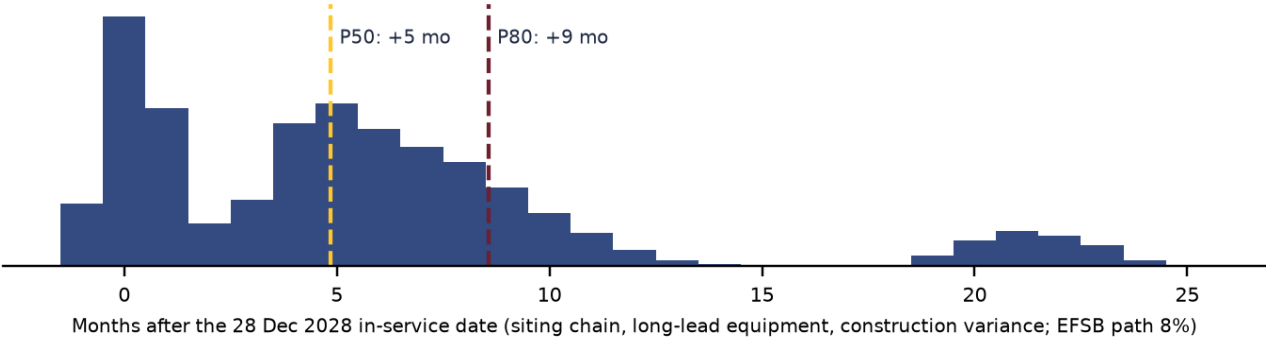
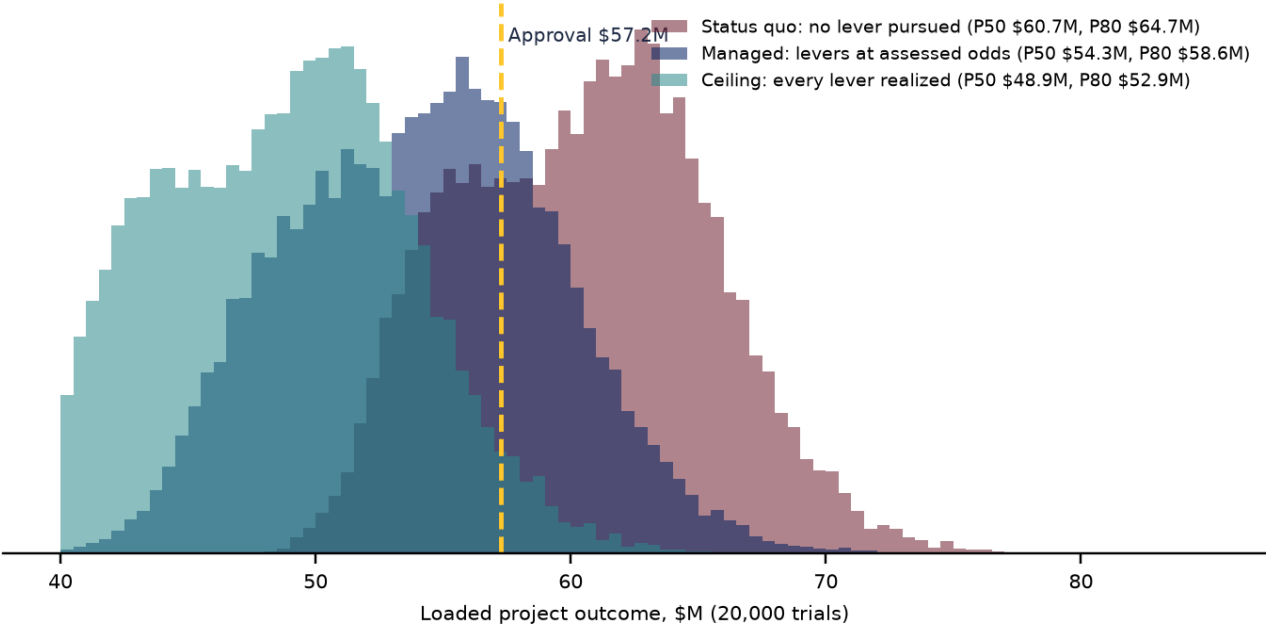
■ **The target is the discipline.** \$25M station-direct is the reference-class 75th percentile for a project this size; the estimate is at \$29.7M. The gap is the scope challenge above, and the PM owns it from the November PAF onward, with the CoE reporting the trend monthly.

Where to be aggressive: two things drive 72% of the variance, and neither is a line item

#	Be aggressive on	Why, from the simulation	Owner	By
1	Close the Haugland reconciliation before the PAF	Threat T1: 58% of the variance in the outcome. A signed scope map turns a +\$3M to +\$12M loaded exposure into a known number; until then every reduction below is gross of it.	PM + Cost Estimating + HEG	Nov 2026
2	Buy siting float	Schedule: 14% of the variance; P50 +5 months, P80 +9 months, each month about \$0.37M. File Article 80 early, set the ZBA hearing within 90 days, pre-brief the abutters on screening and sound.	Siting + PM	Q1 2027
3	Take the decision-only levers now	A1 escalation (\$1.3M expected), A2 risk wording (\$1.1M), B5 disconnect double count (\$0.3M), C3 AFUDC timing (\$0.3M): about \$3M expected, no study needed.	Cost Estimating + Sourcing	Nov 2026
4	Benchmark Station 496 and price its outage windows	B1 (\$0.7M expected) is the largest controllable estimate lever; accuracy on F is 6% of variance; T8 outage premiums are unpriced.	P&C Engineering + EMA	Q1 2027
5	Settle the topology and the GSU count in one study	B2 (\$0.6M expected) and T6 (\$0.4M expected threat) are the same question asked twice. Hosting capacity for two 5 MW feeders decides both.	System Planning + Substation Eng.	Gate 1
6	Hold the civil layer open until geotech	Accuracy on E is 6% of variance and asymmetric (0.6x to 2.6x). B3 is worth \$0.3M expected; the downside is three times that.	Civil + Capital Projects Eng.	Gate 1
7	Get the overhead-basis ruling	A4 (\$0.5M expected at 30% odds, \$2.1M if granted) is the single largest line nobody on the project can change.	Accounting	Nov 2026
8	Price what is missing before someone else does	T2 to T5 (sound, screening, police details, NFPA 855) are \$1.1M expected together; carrying them is cheaper than defending their absence.	Siting + Civil	Nov 2026

Expected values are probability-weighted means from 20,000 trials (mc_scope_schedule.py); 'variance' is the Spearman contribution to the spread of the managed outcome. Reductions in the register total \$10.0 to 14.6M loaded if every one lands; the simulation expects about \$6.3M of it at the assessed odds.

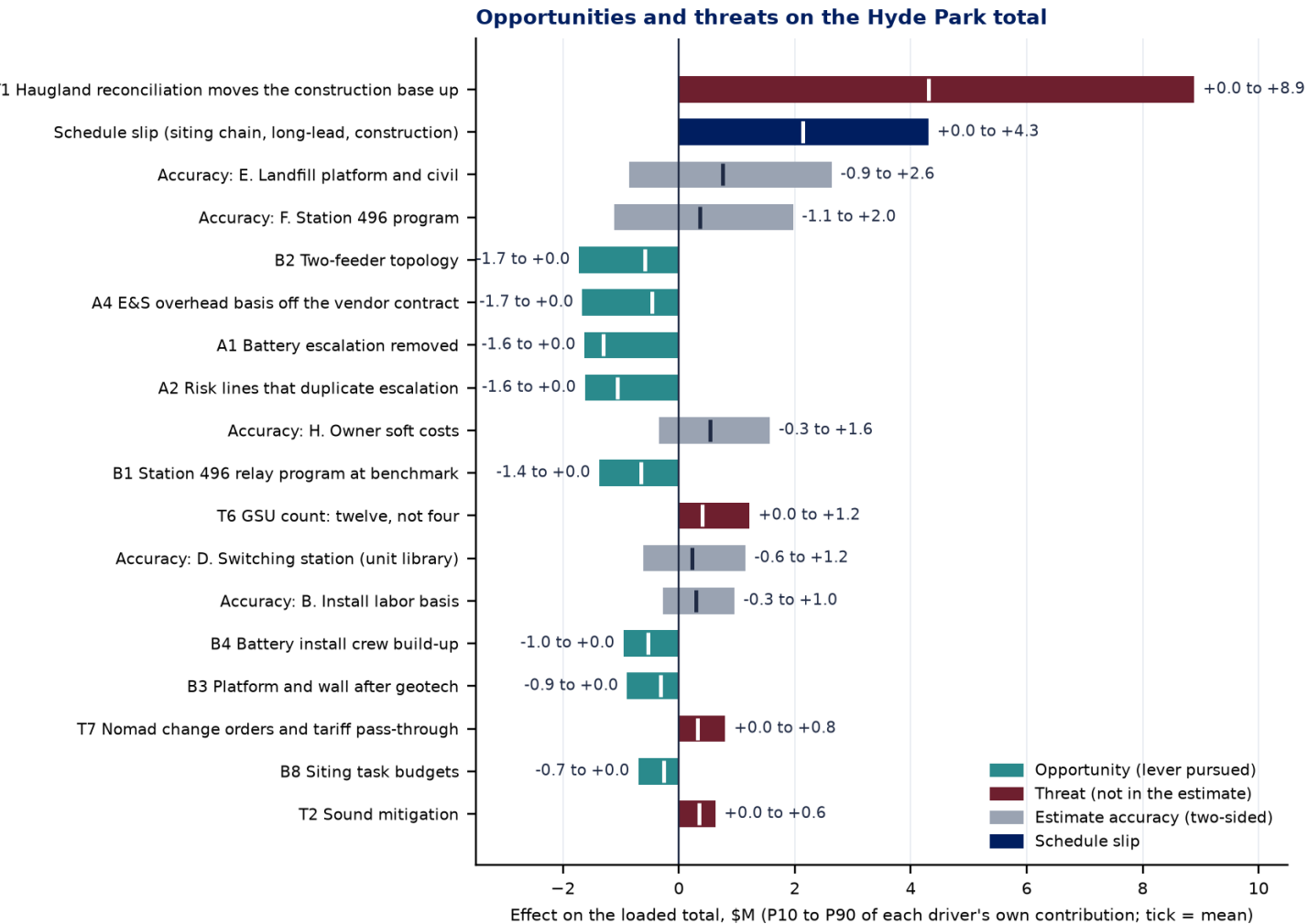
Monte Carlo on cost, scope and schedule: \$61M if nothing is pursued, \$54M if the levers are worked, \$49M if every one lands



Posture	P50	P80	P(over \$57.2M)
Status quo	\$60.7M	\$64.7M	71%
Managed	\$54.3M	\$58.6M	29%
Ceiling	\$48.9M	\$52.9M	4%
Schedule	+5 mo	+9 mo	39% over 6 mo

- **What the three postures mean.** Status quo: the \$57.2M holds in 29% of trials. Managed: 71%, with a 24% chance of landing under \$50M. Ceiling: every lever lands; it is the floor of what is physically possible, not a plan.
- **The reserve.** The \$7.6M of risk register and contingency is what these outcomes draw on. It covers the managed case at about P75 and the status quo at about P25.
- **Schedule.** Eight in ten trials slip; the median slip is 5 months and costs about \$1.8M; an EFSB path (8%) adds 18 to 24 months.

Tornado: the Haugland gap and the siting schedule are the risks; the levers are many, small and mostly ours to take



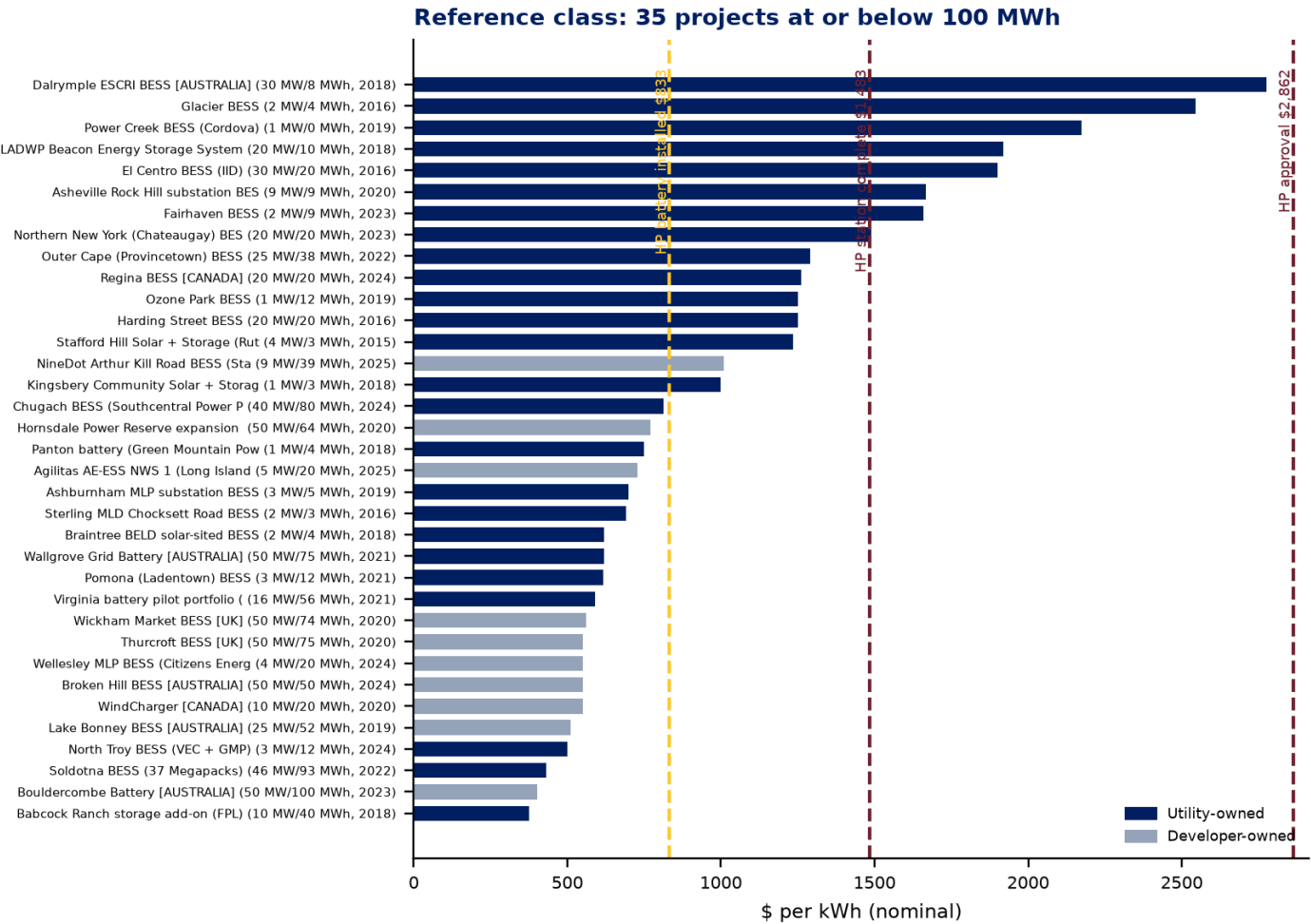
- **Read the right side first.** One threat, the Haugland reconciliation, reaches \$9M at P90 and carries 58% of the variance. Schedule slip reaches \$4.3M. The civil layer and Station 496 are the two accuracy ranges with real upside risk.
- **The left side is a portfolio.** No single lever exceeds \$2M, but together they are worth about \$6.3M at the odds assessed and \$11.8M if every one lands. The four that need only a decision (A1, A2, B5, C3) are the surest.
- **What this says about being aggressive.** Aggression on the station lines buys \$3M to \$6M; aggression on the reconciliation and the siting calendar protects \$8M to \$13M. Do both, in that order.
- **Not on the chart:** the Section 48E credit (\$8.4 to 12.4M), which is larger than everything on the left combined and sits in the revenue requirement.

What must close before this number goes to EPAC as full funding

Item	Type	Dollars at stake	Owner
Reconcile Haugland Class 3 base (\$19.2M, its scope) to Rev 3 layers B+D+E (\$8.3M direct). Red team expects it to move the base up, not down.	Scope boundary	+\$4M to +\$8M direct if half the gap is 21334 scope	Cost Estimating + PM + HEG
Siting chain has no float: ZBA determination Feb 2028, construction start Feb 2028, ISD Dec 2028; EFSB fallback 18-24 mo	Schedule	~\$5M per 12 months (AFUDC, escalation, PM, indirects)	Siting + PM
Present full funding at its real maturity: 30% engineering, no bid, placeholder platform = Class 4	Classification	Range is -25/+50% (\$43M-\$86M) until closed	Cost Estimating
Geotechnical investigation and NFPA 855 hazard mitigation analysis before civil freeze	Study	\$1.0M of deducts vs \$0.6M-\$3.5M exposure	Civil + Capital Projects Eng.
Confirm four written design-basis items (retaining wall, basin, duct bank, GSU count)	Design confirmation	\$1.47M HEG deduct ceiling (~\$0.8M net of GSUs already at 4)	Substation Engineering
Escalation on the fixed-price battery balance; check Exhibit C for tariff / change-in-law pass-through	Decision	\$1.13M	PM / Cost Estimating / Sourcing
Battery installation labor: replace 25%-of-16,801 mh with a crew build-up	Estimate	+/- \$0.5M (Rev 2 used 50%)	Cost Estimating + Nomad + HEG
Station 496 at \$363K per relay: compare to two recent EMA relay-replacement actuals; outage-window premium not priced	Benchmark	Unknown; \$1.8M testing line	P&C Engineering
Spot generation if the BESS misses Dec 2028 (SSF: 2-8 MW) was removed from scope; who carries it?	Scope / ownership	Not in project	System Planning
Re-base D-Line 24211 (2023 conceptual \$4.8M; HEG take-off larger)	Estimate	Unknown	Distribution Engineering

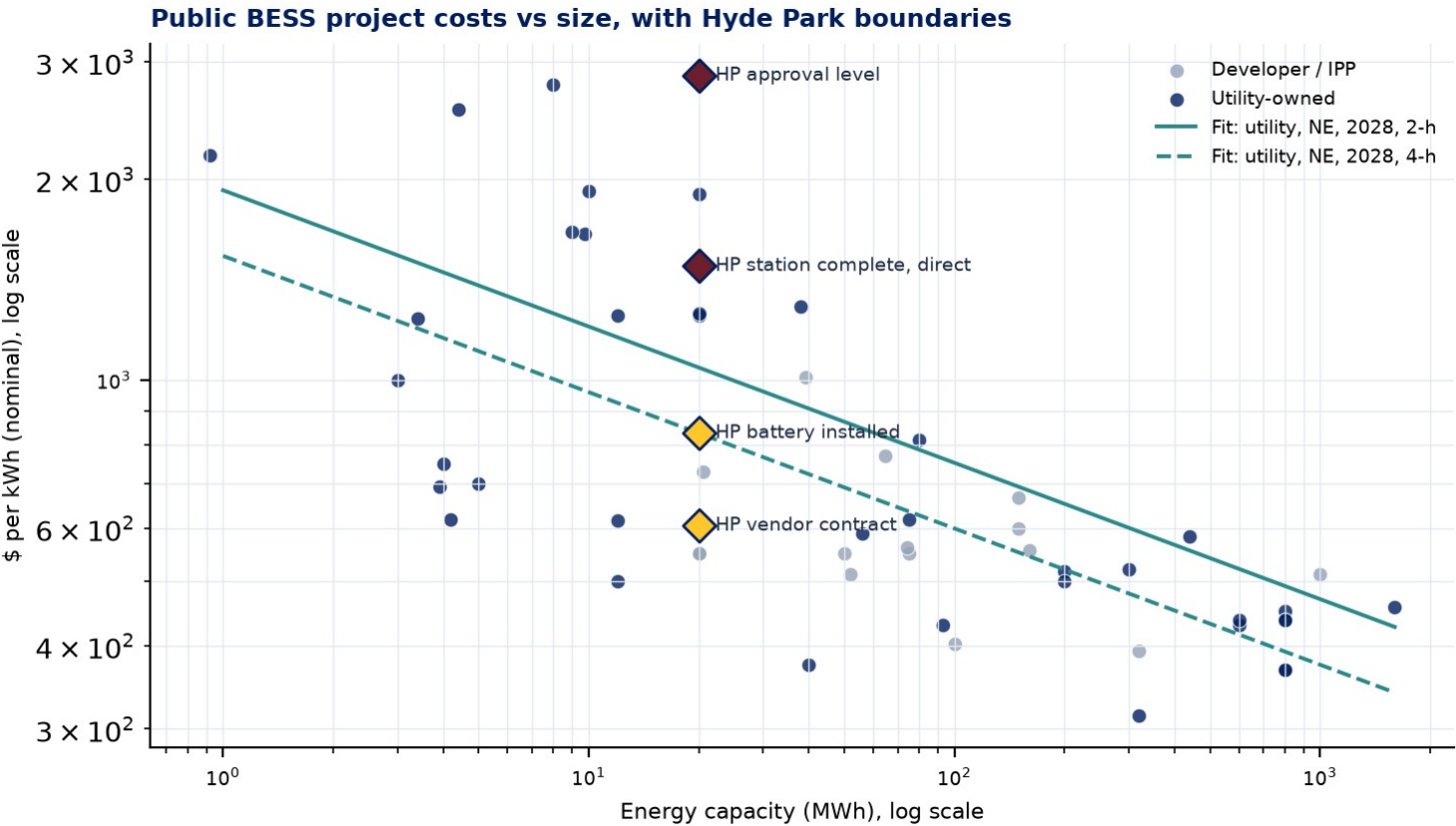
Defensible reductions (~\$3M: battery escalation, design-basis confirmations, material risk sized on non-battery material) are conditional on the Haugland reconciliation closing first.

35 built projects at or below 100 MWh, utility- and developer-owned: where Hyde Park sits



- **Median** \$750/kWh; interquartile \$556 to \$1,275. Utility-owned subclass (25): median \$1,235.
- **Hyde Park battery installed, \$833**: 57% percentile.
- **Station complete, direct, \$1,483**: 77% percentile (68% in the utility subclass), beside Fairhaven (\$1,659 actual, 2023), Chateaugay (\$1,490, 2023) and Provincetown (\$1,289 actual, 2022).
- **Approval level, \$2,862**: above every member. The loaders (\$23.7M) and the two interconnection scopes are what the comparables do not carry.
- **Re-based to 2026 dollars (slides 12 and 14 to 18)**: median \$742 at the installed index, \$816 to \$737 across 30 to 50% anchor battery share (2024 basis, back-cast to each project's year).
- **Built projects only (5 Oct 2026)**. Status of every dataset row verified; 27 rows that were planned, under development, under construction or cancelled are excluded from the class and the regression. 14 members carry a final or actual cost (median \$756); the others are contract, budget or announcement figures for projects that were built.
- **Caveat**: figures from regulator, IDA, EIA and utility documents, verified by status audit; developer-owned members carry developer overhead and profit but no utility indirects or AFUDC; incentives are not netted.

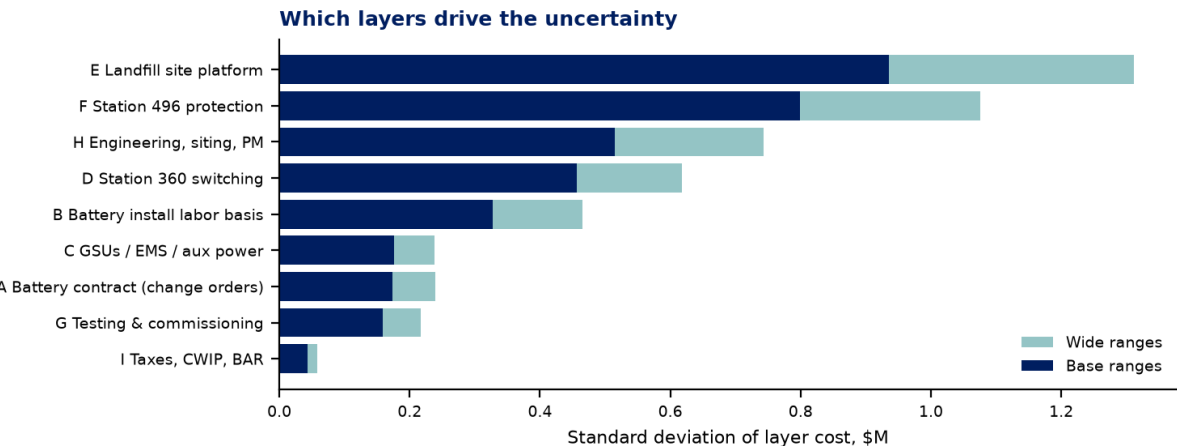
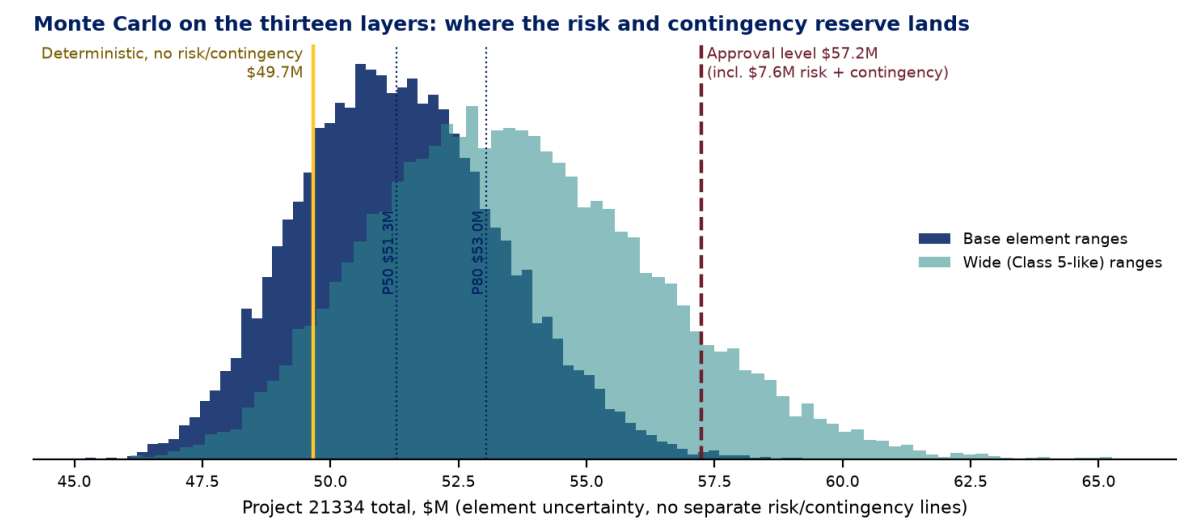
Size and duration explain half the spread in public costs; Hyde Park's direct cost is high-normal



Term	Coefficient	p
Size (ln MWh)	-0.18	0.000
Duration (ln h)	-0.34	0.003
Year	+0.017	0.55
R-squared	0.62	n = 53

- **20 MWh, 2-h, 2028 prediction:** \$968/kWh, 80% interval \$571 to \$1,640.
- **Hyde Park percentiles:** contract 13%, installed 36%, station direct 85%, approval 100%.
- **Use as a bracket, not a price.** Residual spread is a factor of 1.4. In the fuller model utility ownership adds 26% at the same size, duration and year (p 0.01); region is not significant.

Monte Carlo on the layers: reserve covers the modeled ranges, thin against history



	Base ranges	Wide ranges
P50 total	\$51.3M	\$53.4M
P80 total	\$53.0M	\$55.9M
Reserve to reach P80	\$3.4M	\$6.2M
Carried risk + contingency	\$7.6M	\$7.6M
P(total > \$57.2M)	0%	10%

- Distributions** are PERT ranges on each of the nine Station 360 layers plus Station 496, indirects and AFUDC; risk register and contingency are not stacked on top (they fund the same outcomes).
- Estimate-to-actual uplift on comparables:** Fairhaven +64%, Provincetown +9 to +40%, Waena +37%, Regina +31%, Hyde Park Sep cycle +21%. Median about +31%; Rev 3 carries 17.9%.
- Next step:** replace these ranges with the project team's in a two-hour range-estimating session (AACE 41R-08) before full funding.

CONTRIBUTIONS

A \$12 million battery is a \$57 million project, and now every dollar between them has a name

- **Where the money sits.** 292 lines in thirteen layers, reconciled to the penny: battery contract 21%, station and interconnection 32% loaded, owner scope 19%, loaders 41%.
- **Unit cost at every boundary.** \$605/kWh vendor, \$833 installed, \$1,483 station direct, \$2,862 project. Each benchmark read against the row that shares its definition.
- **How it changed.** The battery got cheaper; the project got real. \$43.9M to \$57.2M, five in-service dates, documented from the ESF revisions and the email record.
- **Can it be defended.** Against 35 public projects at or below 100 MWh the station-direct cost is at the 77% percentile and the approval level above every member; a regression on 53 projects brackets a 2028 build at \$571 to \$1,640/kWh; six price curves re-base the class within a tenth of each other. Three kill shots, seven stress tests, a pre-mortem, ten reconciliations.

Decisions requested

- Adopt the layered view (battery, station, interconnection, owner scope, loaders) as the standard presentation of this project, internally and to regulators.
- Make the Haugland-to-Rev 3 reconciliation, the geotech and the hazard mitigation analysis preconditions of the November PAF; present at Class 4 until they close.
- Assign owners to the siting-chain float and the five early-warning indicators.
- Open the Section 48E credit question with Tax now: eligible basis, the FEOC test on the Voyager cells, and the normalization election.
- Adopt the containment plan: a \$1,250 per kWh station-direct target at IFC, five gates with owners, competitive lump-sum pricing of the station and civil scope, and a monthly cost-containment report.